# When should a neural text-entry decoder commit? Risk-controlled commits from a decoder-agnostic trust layer

[TODO: authors, affiliations, e-mail]

Draft v3 (revision round 3) — 12 November 2026 — not for distribution

**Abstract**

*Objective.* Neural text-entry decoders often end in a language model that writes fluent sentences from weak neural evidence, and they rarely signal when a sentence is wrong. We ask when such a system should commit a sentence without confirmation, and whether the rate of wrong commits can be controlled with a stated guarantee. *Approach.* A trust layer reads only the decoder's per-frame output stream, calibrates its confidence and commits, offers alternatives or abstains. The commit threshold is set on held-out calibration data by Learn-then-Test (LTT) and is never lower than the threshold tuned for speed (max rule). Offline, we test it on one intracortical speech participant in a copy task (Brain-to-Text '25), on forearm surface electromyography (sEMG) key presses composed into sentences (19 participants) and on MEG typing (19 participants, negative control), over 50 random fit/calibration/test assignments. Outcomes are wrong commits per sentence attempted, the commit rate, the error among commits and the speed of a simulated gaze-typing user (effective words per minute, eWPM). *Main results.* For the one intracortical participant, always committing sent wrong text for 61% (our decoder) and 19% (published decoder) of sentences. The LTT max rule gave mean wrong-commit rates of 3.0% and 3.1% (target 5%); 8% and 2% of deployments exceeded 5% (allowed: 10%). It committed 15% and 39% of sentences, of which 21% and 8% were wrong, and was 0.11 and 0.51 eWPM faster than the same policy without commits. Conformal risk control met the target only on average: 34–42% of single deployments exceeded it. A guaranteed error among commits of at most 10% was reachable only for the published decoder. On sEMG the layer committed at most 9% of sentences on average, and on MEG none; on MEG and the two weaker sEMG decoders, gaze typing with word completion was faster. Calibrating on data that had also fitted the pipeline raised wrong commits on our decoder (+0.9 percentage points, $p = 0.003$). *Significance.* A stream-level commit rule can keep wrong commits rare with a high-probability guarantee at little speed cost, but with current decoders it commits few sentences, and the evidence comes from one implant participant.

## 1 Introduction

Brain–computer interfaces (BCIs) for text entry are moving towards clinical and consumer use. Intracortical neuroprostheses decode attempted speech, handwriting and typing at high rates [1–5]; non-invasive decoders reconstruct typed sentences from magnetoencephalography (MEG) [6, 7]; and surface electromyography (sEMG) is a candidate input for wearables [8–10]. Most of these systems end in a language-model (LM) stage that turns noisy per-frame predictions into sentences. This lowers the error rate but changes the kind of error: when the neural evidence is weak, the LM still returns a grammatical sentence, and a reader has no cue that it is wrong. In speech recognition, such fluent unsupported output has caused documented harm [11].

For people who depend on the interface to communicate, accuracy is a stated priority: potential BCI users with amyotrophic lateral sclerosis (ALS) or spinal cord injury asked for at least 90% accuracy [12–15], and users of augmentative and alternative communication (AAC) want LM-generated text to reflect their own intent [16, 17]. A sentence sent in their name that

they did not mean is a harm in itself. Current speech neuroprostheses read a finalised sentence aloud without a confirmation step [2] or synthesise voice instantaneously [18], and inner speech can be decoded without an intention to communicate [19]. Whether a decoded sentence should be sent without confirmation (*committed*) is therefore a real design decision.

Existing BCI error handling acts inside one decoder and paradigm (Section 2). Sentence-level LM decoders for speech, handwriting, typing and sEMG now share one structure, per-frame probabilities followed by an LM search. This makes a decoder-independent commit rule possible: a rule that decides whether a sentence is committed, offered for confirmation or not shown, and that controls how often it commits wrong text. The statistical tools exist (calibration, selective classification, conformal risk control and Learn-then-Test [20–24]) and have been applied to single BCI decoders [25, 26] and to LM correction in speech recognition [27]. The methods are not new. What is missing is their use as a commit rule for neural text entry, an evaluation of the guarantee under the way such a rule would be deployed, and the fair comparators: "always offer, never commit" and a no-BCI baseline such as gaze typing [28].

**This work.** We define the problem (Section 3), a trust layer that reads only a per-frame output stream (Sections 4 and 5) and an evaluation protocol (Section 6), and test them offline on public data. Our contributions are:

1. **A stream-only commit rule with a stated guarantee.** The layer calibrates the decoder's per-frame log-probabilities and chooses COMMIT, OFFER or ABSTAIN; the commit threshold is set by Learn-then-Test with a max rule, which bounds the wrong-commit rate with high probability. Algorithms 1–2 give the full procedure.
2. **A test of the guarantee under repeated deployment.** Over 50 random fit/calibration/test assignments we show which guarantees hold per deployment and which only on average, how often a selective target on the error among commits can be met, and that calibrating on data that also fitted the pipeline raises the wrong-commit rate.
3. **A safety-first evaluation protocol**: the wrong-commit rate with the commit rate and the error among commits; always-commit, always-offer, gaze-typing and word-completion baselines; a simulated-user speed metric with a sensitivity analysis; and signal-destroying and intent-mismatch controls.

The output format (NeuralTextStream v0.1), its adapters, an audit log and all scripts are released under Apache-2.0 (Appendix L).

Figure 1 summarises the result. The commit decision was tested non-trivially on one intracortical participant, where the layer turned frequent wrong commits into rare ones; on sEMG it committed few composed sentences, and on MEG it correctly never committed (a negative control). Its speed gains were small and user-model dependent, so our claim is about safety: with the guaranteed rule, the wrong-commit rate stayed at or below 5

## 2 Related work

**Brain-to-text decoders.** MEG decoding of typed sentences reached a CER of 0.29 with a keystroke-locked (synchronous) model [6]; a second version decodes without key times (asynchronous) and adds an LLM stage [7]. Other non-invasive work decodes perceived speech or words [29–31]. Intracortical decoders reach high accuracy for handwriting, speech and typing [1, 2, 4, 5, 32], with public phoneme logits [33, 34]; sEMG text entry exists for the wrist and forearm [8–10]. Most use CTC [35] with beam search and an LM [36]. In the intracortical study whose data we use, decoded words were displayed while the participant spoke; a sentence was finalised after 6 s without speech or when he selected an on-screen button by eye gaze, it was then read aloud by text-to-speech, and he rated it afterwards as correct or not; correctly

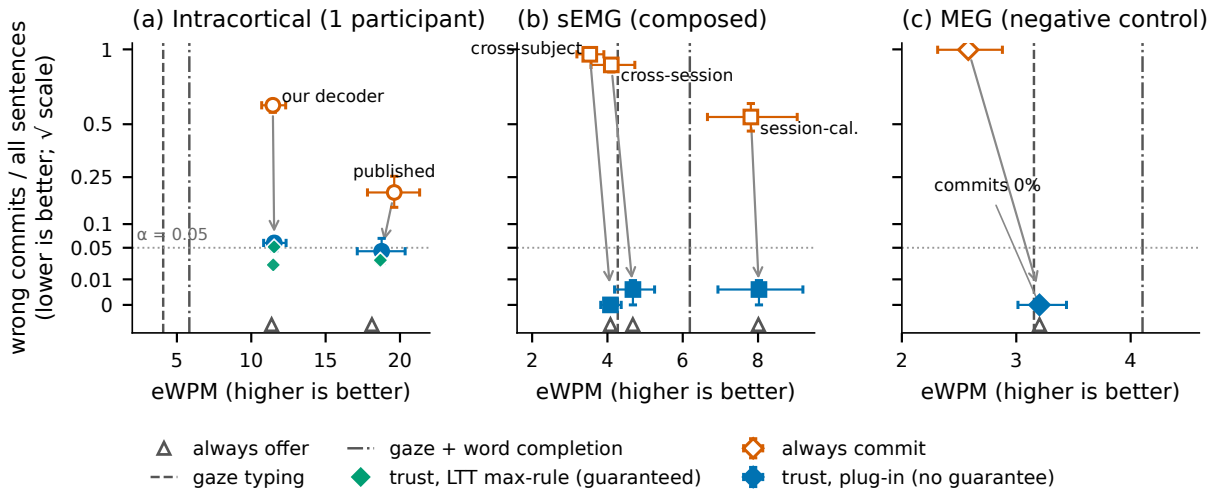



**Figure 1: The trust layer turns frequent wrong commits into rare ones; on MEG it does so by never committing.** Wrong commits per sentence attempted (signed square-root axis) against effective words per minute (eWPM) of a simulated user. Open markers: always commit the top sentence; blue: the plug-in trust layer on the same output (arrows), which has no guarantee; green diamonds: the guaranteed rule (fit-A pipeline, clean calibration, LTT max rule), deployment splits A and B; triangles: always offer (no wrong commits; drawn just below 0). Dashed line: gaze typing; dash-dotted line: gaze typing with word completion; dotted line: $\alpha = 0.05$. Error bars: 95% cluster-bootstrap CIs over days (a), participants (b) and sentences × participants (c), conditional on the fitted pipeline. (a) Intracortical speech (Brain-to-Text '25, one participant, 816 test trials): our phoneme-to-word decoder (plug-in: 23% committed, error among commits 0.25; guaranteed: 10% and 19%, errors 0.24 and 0.27) and the published sentence (plug-in: 47%, 0.09; guaranteed: 40% and 42%, 0.07). (b) Forearm sEMG, 551 composed sentences, 19 participants, three decoder conditions; the guaranteed rule (not drawn) gives the same commit and wrong-commit rates as the plug-in rule. (c) MEG typing, 637 instances of 17 sentences, 19 participants; the trust layer equals always offer.

rated sentences fine-tuned the decoder online [2]. There was thus no confirmation before text-to-speech in that interface, which corresponds to our COMMIT (a later interface version let the participant control when text-to-speech played). Other speech neuroprostheses make the same decision implicitly [3, 18, 37].

**Error handling in BCIs.** P300 and RSVP spellers commit a character when its posterior passes a threshold and stop early when confident [38–40], often with an LM prior [41–43]. Error-related potentials allow automatic error detection [44–48]; asynchronous BCIs detect a "no-control" state [49–51]; myoelectric control rejects uncertain decisions [52]; adaptive decoders change the decoder over days [53–57]. These rules are built into one decoder and paradigm; ours reads only the decoder's per-frame output, leaves the decoder fixed, and states a finite-sample bound.

**Calibration, selective prediction and risk control.** Temperature scaling fixes over-confidence [20] but degrades under shift [58]; the maximum softmax probability detects errors [59]; selective classification trades coverage for risk, and selection with guaranteed risk (SGR) bounds the error among accepted items with high probability [21]; risk-controlling prediction sets give the corresponding high-probability bound for set-valued predictions [22]. In BCI, confidence-aware abstention [25] and calibrated alternatives [26] were shown for single decoders; dialogue systems accept or reject utterances by cost-based thresholds [60]. Conformal risk control (CRC) bounds the expected value of a monotone loss [23, 61], Learn-then-Test (LTT) bounds the risk with high probability [24], weighted variants give approximate bounds under shift [62–64], and online methods give long-run guarantees for arbitrary sequences [65–68].

We use these methods as published; our contribution is their use and evaluation as a commit rule. Conformal methods were applied to speech recognition [27, 69] and to LM outputs and abstention [70–72], where LM stages can produce fluent unsupported text [73, 74].

**Metrics and standards.** Utility metrics [75], the practical bit rate [76] and text-entry metrics [77] charge for corrections, unlike information transfer rate [78]; LM prediction speeds up gaze typing [28, 79], although scanning prediction lists has a cognitive cost that can offset the saved keystrokes [80]. Our eWPM is close to the utility metric. NWB [81], BCI2000 and OpenViBE [82, 83], IEEE P2731 and P2794 [84–86] and decoder benchmarks [87, 88] each cover part of the stack; none defines a per-frame decoded-output format with confidence semantics.

## 3 Problem setting

**Items, streams and actions.** An *item* is one attempt to enter one sentence: a *trial* of the intracortical data, an *instance* (one participant typing one sentence) of the MEG data, or a composed *sentence* of the sEMG data. For item $i$, the decoder emits a stream of $F_i$ frames of log-probabilities $\log p_t(v)$ over $V$ tokens. An LM search returns $K = 10$ ranked candidate sentences with scores (the published-sentence gate of Section 5 ranks 11: the published sentence and our 10), from which we compute $\hat{p}_{\text{top1}}$, the calibrated probability of the top candidate. The system then takes one action: COMMIT the top candidate (sent without further input); OFFER up to $k$ candidates for the user to pick; or ABSTAIN, after which the user types with a fallback keyboard (an OFFER_WORDS action, which marks uncertain words, appears only in the ablation). Let $c_i \in \{0, 1\}$ indicate that the top candidate differs from the intended sentence (exact match after normalising case and punctuation); for committed items, the simulated user's wrong commit equals $c_i$.

**The two error quantities.** Over the $n_{\text{test}}$ test items,

$$\text{wrong-commit rate} = \#\{\text{wrong commits}\}/n_{\text{test}} \approx \Pr(\text{commit} \wedge \text{wrong}), \tag{1}$$

$$\text{error among commits (EAC)} = \#\{\text{wrong commits}\}/\#\{\text{commits}\} \approx \Pr(\text{wrong} \mid \text{commit}). \tag{2}$$

The first is the *marginal* risk: how often wrong sentences reach the recipient per sentence attempted. It is monotone in the commit threshold, so CRC and LTT can bound it (Section 5.3). Committing rarely makes it small, so we always report it with the *commit rate*. The second is the *selective* risk that a user sees in committed text. We report it next to every result and test controllers that bound it (Section 5.4). Table 4 (Appendix A) collects the symbols; $\alpha = 0.05$ is the target wrong-commit rate, $\beta$ the target error among commits and $\delta = 0.1$ the failure probability of the high-probability guarantees.

## 4 The stream interface

The trust layer reads one output stream per item (architecture: Figure 5 in Appendix L). In our open format, NeuralTextStream (NTS) v0.1, a stream of $F$ frames over $V$ tokens holds an $F \times V$ array of natural-log probabilities, the frame times, the ordered vocabulary with its type (`char`, `key`, `phoneme`, ...) and the CTC blank index. Two rules matter for the commit decision: a producer must emit the decoder's own, unmodified output distribution (calibration is left to the consumer, which can fit it on data with known truth), and ground-truth text must not appear in the stream. One small adapter per device maps decoder output into the stream (devices and data: Table 13 in Appendix L); every later step is the same code for all devices, with parameters re-fitted per device (Table 8). Appendix L describes the serialisations, the conformance suite, the adapters and the audit log.

# 5 The trust layer

The trust layer reads one stream per item and returns one action. It never reads raw signals or decoder internals. Algorithm 1 (Appendix B) gives the decision for one item and Algorithm 2 the fitting and calibration; Table 8 lists the fitted values and the policy grid. All parameters are fitted per device on that device's fit data.

## 5.1 Calibration

Decoder log-probabilities are often over-confident. We fit a frame temperature $\tau_f$ that rescales every frame, $\tilde{p}_t(v) \propto p_t(v)^{1/\tau_f}$. Its fitting frames are the *greedy CTC peaks*: the centre frame of each run of identical non-blank arg-max tokens; each peak is aligned to the true text by a Levenshtein alignment of the greedy text, and its target is the aligned true token (the blank for an inserted peak). $\tau_f$ minimises the cross-entropy of these targets. We then fit a sentence temperature $\tau_s$ on the scores of the $K$-best list, $\hat{p}_{\text{top1}} = \text{softmax}(\text{scores}/\tau_s)_1$, by minimising the binary negative log-likelihood of "the top candidate is right" [20]. Both are golden-section searches over $\log \tau \in [\log 0.05, \log 20]$; $\tau_s$ reached the upper end (20, which flattens the confidence) on MEG and sEMG cross-subject. The frame temperature is applied *before* the LM search, so it changes the decoded text as well as the confidence (Section 7.6).

## 5.2 Candidates, guard and policy

Candidates come from a CTC prefix beam search with a character 6-gram LM trained on Tatoeba sentences [35, 36]. For intracortical speech, a phoneme beam search is followed by a CMUdict lexicon, a Tatoeba word unigram and the same character LM. Each character of the top candidate gets an acoustic confidence: on character streams, the largest calibrated frame posterior of that character over its frames in a CTC forced alignment; on the phoneme path, the lowest frame posterior of the word's phonemes times $0.5^e$, where $e$ is the number of phoneme edits needed to match the lexicon. An optional *guard* scores how much of the top candidate the LM invented: the larger of the share of characters with confidence below 0.5 and one minus the posterior-weighted word agreement between the top candidate and the top five (formulas in Algorithm 1). A split-conformal sentence set (score $1-\hat{p}$, level 0.1, calibrated on fit items whose truth is in the $K$-best list) gives the candidates that an OFFER shows.

The policy takes, in this order: ABSTAIN if the guard risk is at least $r_{\text{abst}}$; COMMIT if $\hat{p}_{\text{top1}} \geq \lambda$ and the guard risk is at most $r_{\max}$; OFFER the conformal set if it has between 1 and $k_{\max}$ candidates; otherwise the fallback (offer the top $k_{\max}$, or abstain). Its thresholds are chosen on fit data from a grid (Table 8) to maximise eWPM (Section 6) subject to a wrong-commit rate of at most $\alpha$ on the fit data (first maximum in grid order). This *plug-in* fit has no guarantee on new data. The guard was part of the configuration fixed before the evaluation runs; it raised wrong commits on intracortical data and had no effect elsewhere (Section 7.6).

**Gate on a published sentence.** For the published intracortical decoder we have the stream and the published LM-rescored sentence, but not the decoder's LM or weights. We put the published sentence on top of the $K$-best list of our own phoneme-to-word decoder, with a score equal to our best score plus a fixed bonus of 2.0, and give each published word the acoustic confidence of the aligned word in our top candidate (0 if no word aligns). The guard and the policy then run unchanged.

## 5.3 Finite-sample control of the wrong-commit rate

The commit score $s_i$ is $\hat{p}_{\text{top1}}$ if the other policy rules allow a commit (guard risk $< r_{\text{abst}}$ and $\leq r_{\max}$), and $-\infty$ otherwise. The layer commits when $s_i \geq \lambda$, with loss $L_i(\lambda) = c_i\,\mathbf{1}[s_i \geq \lambda]$ (the

non-commit actions have loss 0). $L_i$ is non-increasing in $\lambda$, and "never commit" ($\lambda = +\infty$) has loss 0. On $n$ calibration items with known truth and empirical risk $\hat{R}_n(\lambda)$:

- **CRC** [23] chooses $\hat{\lambda}_{\text{CRC}} = \inf\{\lambda : \frac{n}{n+1}\hat{R}_n(\lambda) + \frac{1}{n+1} \leq \alpha\}$ over the calibration scores and $+\infty$. If the calibration items and a new item are exchangeable, $\mathbb{E}[L_{n+1}(\hat{\lambda}_{\text{CRC}})] \leq \alpha$, where the expectation is over the calibration draw and the new item. It bounds the mean over deployments, not a single deployment. For $n < 1/\alpha - 1 = 19$ it never commits.
- **LTT** [24] tests $H_0(\lambda)$: $R(\lambda) > \alpha$ along a fixed sequence from $\lambda = 1.00$ down to 0.00 in steps of 0.01, with the exact binomial p-value $\Pr(\text{Bin}(n, \alpha) \leq n\hat{R}_n(\lambda))$, and stops at the first non-rejection; $\hat{\lambda}_{\text{LTT}}$ is the last rejected $\lambda$. If the calibration items are i.i.d. from the deployment distribution, $\Pr\big(R(\hat{\lambda}_{\text{LTT}}) \leq \alpha\big) \geq 1-\delta$, where $R$ is the population risk at the chosen threshold. This is a per-deployment statement. LTT cannot commit below $n = 45$ ($0.95^{45} < \delta$).
- **Max rule.** The deployed threshold is $\max(\lambda_{\text{tuned}}, \hat{\lambda})$, where $\lambda_{\text{tuned}}$ is the plug-in threshold. Because $L_i$ is non-increasing, a larger threshold has a pointwise smaller loss, so both guarantees carry over. This holds when $\lambda_{\text{tuned}}$ is fixed before calibration and also when it is tuned on the calibration data subject to $\lambda \geq \hat{\lambda}$. "Pure" CRC or LTT drop $\lambda_{\text{tuned}}$ and fill the risk budget.

**Our primary rule is LTT with the max rule, calibrated on clean data.** We chose it in this revision because the per-deployment guarantee is the one a user experiences, and because the repeated deployments (Section 7.2) showed that CRC and the plug-in exceed $\alpha$ in a third or more of single deployments. CRC with the max rule is reported next to it. The implementation commits when $s \geq \lambda$ (left-continuous loss); the committed set is a subset of the set $\{s > \lambda'\}$ for $\lambda' < \lambda$, so the bounds stay valid and slightly conservative.

### 5.4 Controllers for the error among commits

A user sees the error among commits, which the controllers above do not bound. We test three controllers with target $\Pr(\text{wrong} \mid \text{commit}) \leq \beta$ with probability at least $1 - \delta$, for $\beta \in \{0.05, 0.10, 0.20\}$. *SGR* [21] searches the sorted scores of the eligible calibration items by bisection and accepts the lowest threshold whose exact binomial upper bound on the calibration error among commits is at most $\beta$, at level $\delta/\lceil \log_2 m \rceil$ for $m$ eligible items. *LTT-sel* tests $H_0(\lambda)$: $\Pr(\text{wrong} \mid s \geq \lambda) > \beta$ with the exact binomial p-value given the number of calibration commits, with a Bonferroni correction over the grid $\lambda = 0.05, 0.10, \ldots, 1.00$. *Joint* uses $\max(\lambda_{\text{SGR}}, \lambda_{\text{CRC max}})$, which targets both the marginal and the selective risk. A selective bound conditions on committing, so it needs many calibration commits.

**Separate data.** Both guarantees need the parts that define $s_i$ to be fitted on data disjoint from the calibration items. The deployment protocol (Section 6) ensures this, with three exceptions listed in Appendix N: the Brain-to-Text '25 word-decoder grid was tuned on fit trials that overlap the calibration set (re-tuning on fit-A selected the same setting), the MEG checkpoint was selected on validation data that include the calibration sentences, and sEMG session-calibrated uses two cross-fitted decoders for calibration and test.

### 5.5 Online re-calibration as a safeguard

Exchangeability can fail during use. An online controller re-tunes the commit threshold from user feedback, with a long-run guarantee for any sequence of items (`aci`: $\lambda_{t+1} = \max(0, \lambda_t + \eta(L_t - \alpha))$, $\eta = 0.05$ [65, 67, 68]; variants in Table 14, Appendix M). Its bound says nothing about short windows and does not see a wrong commit that the user does not notice (a *silent error*).

# 6 Evaluation protocol

## 6.1 Outcomes, user model and baselines

The primary safety outcome is the wrong-commit rate (Section 3), always reported with the commit rate and the error among commits. Speed is secondary. We simulate a user who reads the output, accepts or corrects a commit, picks from an offer, or types the sentence with a fallback keyboard. For item $i$ with intended sentence $y_i$, the simulation returns the total time $t_i$ (input, reading, selections, corrections, retyping) and the final text, and

$$\text{eWPM} = 60 \cdot \frac{\sum_i \mathbf{1}[\text{final text}_i = y_i]\ \text{words}(y_i)}{\sum_i t_i}. \tag{3}$$

Like the BCI utility metric [75], eWPM charges the time needed to fix errors. A *silent error* is a wrong final text that the user did not correct; it counts zero words, and we report its rate separately because eWPM does not price its harm.

The main user types with a gaze keyboard at 20 characters per minute and scans and selects one of $k = 3$ word alternatives in 4.5 s. Both values are matched to one eye-gaze user with ALS in a non-BCI setting [28]; speeds vary widely across gaze users, and all other costs are assumptions (Table 5). The neural input runs at 159.3 (intracortical), 75.8 (sEMG, median press period) and 152 (MEG) characters per minute; the intracortical rate comes from copy-task speech durations, and free communication will be slower because it adds planning time. After a COMMIT, the user misses each wrong word with probability 0.05, which makes a silent error; after an OFFER, the main user sees every error and chooses the cheapest of picking, fixing or retyping. This asymmetry favours offers and is not measured. For the able-bodied MEG and sEMG typists, eWPM is hypothetical. Because only two costs are measured, we refit every policy under eight variants (fallback 10 or 40 characters per minute, selection costs $\times 0.5$ or $\times 2$, miss probability 0 or 0.2, and a *symmetric-offer* user who also misses wrong words in offers, with miss probability 0.05 or 0.2; Appendix D).

*Baselines:* always commit the top candidate; always offer and never commit, the fair comparator for a commit rule because the layer can gain speed only by committing; a *plain threshold* on the calibrated $\hat{p}_{\text{top1}}$ without the guard; gaze typing alone (error-free, no reading time, which favours it); and gaze typing with *word completion*, where the keyboard shows the three most frequent words with the typed prefix (a context-free unigram without the test sentences) and the user selects the intended word when this is faster. Clinical AAC uses stronger, context-dependent prediction (n-gram or LLM), so a realistic AAC baseline would probably be faster than ours; on the other hand, our main completion baseline charges no time for looking at the list, which favours it [80].

## 6.2 Datasets

**Intracortical speech (Brain-to-Text '25).** The public copy-task file holds the online phoneme CTC logits, cue sentences and published LM-rescored sentences of participant T15, an implant recipient in the BrainGate2 clinical trial [2, 34]: 1,718 trials from 15 post-implant days. We use the 1,628 trials with the 125,000-word vocabulary (14 days); the 90 trials with the 50-word vocabulary (all of day 25 and 40 of day 27) are a different task and are excluded. The file has no time stamps, so the file order within each day is the trial order. The main pipeline is fitted on the first half of each day (812 trials) and tested on the second half (816). We evaluate our phoneme-to-word decoder (test WER 0.149) and the published sentence (WER 0.043), gated as in Section 5.

**Forearm sEMG (Toronto).** The dataset holds isolated key presses from 19 participants over two days [9, 89]. We compose held-out Tatoeba sentences from single presses with oracle timing

and no inter-key context (570 fit, 551 test sentences). A press can recur across test sentences, so participant intervals are primary. A CNN classifies each press under three conditions: *session-calibrated* (day 1 plus half of the day-2 recordings, cross-fitted), *cross-session* (day 1 only) and *cross-subject* (leave-participants-out). Half A of the fit sentences fits the temperatures, the conformal sets and the LM weight, half B the policy.

**MEG typing (SpanishBCBL).** Healthy participants typed memorised Spanish sentences in an MEG scanner [6, 90]. We split 19 participants' data by a hash of the sentence text into 101 training, 11 validation and 17 test sentences (637 test instances); the same people appear in all splits, and no test sentence is in training or LM text. All trust parameters are fitted on the 437 validation instances. Our asynchronous ConvCTC decoder (Appendix K) gets no key times at inference, but it uses key times as training labels and sentence bounds from the keyboard log. The trust harness uses the beam settings of the earlier harness (insertion bonus 1.0); with the validation-tuned beam the layer also committed nothing (3.14 instead of 3.20 eWPM).

### 6.3 Calibration data and deployment splits

The *deployment protocol* calibrates on data that played no other role. On Brain-to-Text '25, each day's fit trials are split into fit-A (temperatures, conformal sets, guard and policy; 411 trials) and a clean calibration set (401 trials), by alternating trials (split A) or by taking the earlier half as fit-A (split B); both are tested on the same 816 test trials. On sEMG, a random half of fit half B fits the policy grid (142 sentences) and the other 143 only calibrate; on MEG, 6 validation sentences (238 instances) fit the pipeline and 5 (199 instances) calibrate.

**Repeated deployment splits.** Two fixed splits do not show how a guarantee behaves over deployments. We therefore repeat the protocol 50 times. On Brain-to-Text '25, each repeat permutes every day's 1,628 trials and splits them day-stratified into fit-A (1/4, about 405 trials), clean calibration (1/4) and test (1/2, about 818); the whole pipeline is fitted on fit-A. On sEMG, each repeat splits every participant's fit sentences into fit-A (1/2: temperatures, conformal sets), fit-B1 (1/4: policy grid) and fit-B2 (1/4: clean calibration), with the fixed 551 test sentences. For each repeat we record the test rates of the plug-in rule, pure CRC, CRC and LTT with the max rule, and the selective controllers, and, as a *reuse* condition, pure CRC calibrated on data that also fitted the pipeline (fit-A on Brain-to-Text '25, fit-B1 on sEMG), paired with the clean calibration of the same repeat. For speed, the fitted frame temperature was rounded to a step of 0.02 (Brain-to-Text '25) or 0.01 (sEMG) so that decodes could be cached; the largest change was 0.27%. Repeats re-use one finite data set per device, so their spread is split-to-split variation for these participants, not a confidence interval for new data. The test half is a proxy for the population risk, so test-sampling noise (standard error about 0.008) adds to the share of repeats above $\alpha$.

**Random splits and the exchangeable reference.** Random calibration/test splits of the test pool are exchangeable by construction and check the implementation (Appendix Q). As a reference for the fixed deployment splits, we pool the calibration and test items and draw 1,000 random splits of the same sizes (for Brain-to-Text '25 also within days); the one-sided $p$-value is $(k+1)/(B+1)$, where $k$ of $B$ reference splits have a test rate at least as high as observed.

### 6.4 Controls and statistics

Decoder controls (shuffled MEG, zero input, permuted training labels) test that the decoders use the signal. *Signal-destroying* trust controls keep the stream format but remove the neural

information (frame-shuffled and near-uniform streams, zero-input output, a permuted-label decoder); a stream-only layer should not commit these. The *intent-mismatch* control is the real stream of a different sentence by the same participant, which a stream-only layer cannot detect.

All confidence intervals come from one bootstrap: 95% percentile intervals, $B = 2000$, seed 20261112, resampled at the primary unit (days for Brain-to-Text '25, 14 clusters; participants for sEMG, 19; sentences and participants at once for MEG, two-way), so one quantity has one interval in every table. They are conditional on the fitted pipeline and the threshold: they describe test-set variation, not fit or calibration variation, which the repeated splits show. Because 14 clusters of unequal size are few for a percentile bootstrap, we add exact day-level sign tests for the key paired differences. For rates with no events we give exact one-sided 95% Clopper–Pearson upper bounds (items treated as independent, which is optimistic under clustering), and for fewer than 30 commits we give the error among commits as counts. The six tests of the fixed deployment splits against the exchangeable reference are Holm-corrected. The primary endpoints (wrong-commit rate of the trust layer and the paired eWPM difference to gaze typing) were named in revision 1, after the first evaluation runs; the always-offer and word-completion baselines, the user-model variants, the ablation, the deployment protocol, the repeated splits, the selective controllers and the choice of the LTT max rule as the primary row were added in later revisions. All inference is therefore exploratory. All decoder settings, user-model values, the policy grid, $\alpha$, $\beta$, $\delta$ and the online step sizes were fixed before the runs that use them.

# 7 Results

## 7.1 The decoders use the neural signal

Each decoder reads its input (Table 6 in Appendix E). Our asynchronous MEG decoder reached a test CER of 0.506 (greedy) and 0.475 (beam and LM tuned on validation); shuffled MEG gave 0.760 and zero input 0.925, and a text-only output that ignores the MEG reaches 0.688. Over three training seeds, the test CER was $0.533 \pm 0.051$ (greedy) and $0.503 \pm 0.056$ (beam and LM); seed 2 stopped early at a worse validation CER and was worse on test (0.592 and 0.567), and all seeds stayed at least 0.14 below the shuffled control ($0.746 \pm 0.014$; Table 7). In the trust harness the top-1 sentence was right for 2 of 637 instances. For context, the published asynchronous encoder reports $0.59 \pm 0.02$ with another split and without key-time training labels [7]. On Brain-to-Text '25, our phoneme-to-word decoder reached WER 0.149 with the right sentence first for 39% of test trials, and the published decoder WER 0.043 with 81%. On sEMG, balanced key accuracies (0.367, 0.240, 0.226) were well above uniform chance (0.037; permuted labels 0.032), and after the LM the right sentence was first for 46%, 12% and 4% of composed sentences.

## 7.2 The guaranteed rule keeps wrong commits rare in most deployments

Table 1 gives the policies on the fixed test sets and Table 2 the repeated deployments.

**Intracortical speech, one participant.** Always committing sent wrong text for 0.610 of trials on our decoder and 0.194 on the published sentence, whose WER is only 0.043, because one wrong word makes a sentence wrong. Over 50 repeated deployments, the guaranteed rule (LTT max rule, clean calibration) gave a mean wrong-commit rate of 0.030 (5th–95th percentile over deployments 0.016–0.052) on our decoder and 0.031 (0.020–0.047) on the published gate; 8% and 2% of deployments were above $\alpha = 0.05$, within $\delta = 0.1$ (Table 2). It committed 15% and 39% of trials, with an error among commits of 0.21 and 0.08, and was 0.11 (0.06–0.16) and 0.51 (0.44–0.60) eWPM faster than the same fitted policy without commits, in every deployment. Its cost against CRC with the max rule was 0.010 and 0.036 eWPM. Pure CRC met its own target, the mean: 0.046 and 0.049 (Monte-Carlo standard error about 0.002). But 34% and 42%

of single deployments were above $\alpha$, as were 34% and 22% for the plug-in rule fitted on the same data. On the fixed test set, the guaranteed rule gave 0.025 and 0.052 (deployment splits A and B) on our decoder and 0.029 and 0.031 on the published gate (Table 1).

The plug-in trust layer fitted on all 812 fit trials, which has no guarantee, gave 0.059 [0.045, 0.070] on our decoder, above $\alpha$; 48 of its 189 commits were wrong. The difference to the fit-A rows does not come from CRC. In the two fixed splits, the fall from 0.059 to 0.025 (split A) came entirely from the policy-grid choice on fit-A (commit threshold 0.3 and maximum guard risk 0.1, against 0.5 and 0.5 on the full fit data): the full-fit temperatures with the fit-A policy also gave 0.0245, and CRC changed nothing. In split A, CRC was not binding (its calibration bound at the chosen threshold was 0.0075); in split B, the fit-A policy (threshold 0.4, guard 0.3) gave 0.060, and CRC removed 4 correct commits. Split A is not typical: over the repeated deployments the CRC max rule changed the plug-in decisions in 54% (ours) and 34% (published) of deployments, and the plug-in operating point varied widely (commit rate 0.13–0.26, wrong-commit rate 0.021–0.072 on our decoder). Per day, the plug-in layer beat always offering on 12 of 13 days with a difference (ours; sign test $p = 0.003$) and 14 of 14 (published; $p = 0.0001$); against always committing it tied (7 of 14 days; $p = 1.0$) and lost on 13 of 14 days (published).

**Forearm sEMG (composed sentences).** The layer cut the wrong-commit rate from 0.541, 0.882 and 0.962 to 0.004, 0.004 and 0, committing 21, 10 and 0 of 551 sentences (Table 1); the guaranteed rule gave the same rows. Over repeated deployments it committed 9%, 2% and 0% on average, with mean wrong-commit rates of 0.011, 0.006 and 0, and never above $\alpha$ (Table 2). Only the session-calibrated decoder clearly beat gaze typing [+2.71, +4.91] (faster for 18 of 19 participants) and gaze typing with word completion [+0.82, +2.99]; for the cross-session and cross-subject decoders, the difference to gaze typing was [−0.07, +0.97] and [−0.44, +0.09], and word completion was faster than the layer. These speeds depend on templated sentences: 295 of the 551 test sentences have a near variant in the LM training text. On the other 256, trust eWPM fell from 8.02 to 6.97, from 4.68 to 4.08 and from 4.08 to 3.69, always-commit wrong-commit rates rose to 0.664, 0.961 and 0.992, and on the cross-subject decoder gaze typing was faster than the layer [−0.66, −0.23]; the layer still committed almost nothing (1 wrong commit in 256 sentences).

**MEG (negative control for the commit decision).** The intended sentence was in the 10-best list for 1.6% of instances, and always committing gave a wrong-commit rate of 0.997. The trust layer committed nothing (0 of 637, one-sided 95% upper bound 0.0047), so it equals always offering, on all three seeds. Its difference to gaze typing was [−0.10, +0.25] two-way (sentences [−0.04, +0.15], participants [−0.08, +0.22]) on seed 0, about zero on seed 1, and negative on the weaker seed 2 [−0.30, −0.05]; gaze typing with word completion was faster [−1.13, −0.64]. No guarantee for new sentences is possible on these data: instances of one sentence are dependent, so the exchangeable unit is the sentence, and a sentence-level CRC needs at least 19 calibration sentences (LTT 45), while the clean calibration set has 5 and the whole test pool 17. With calibration and test sentences disjoint, instance-level CRC gave mean test rates of 0.016–0.019 (95th percentile at most 0.038; never above $\alpha$ in 1,000 splits each with 5, 8 or 11 calibration sentences), but this is a property of the decoder: only 2.4% of instances were eligible to commit, and 0.91–0.94 of the commits that pure CRC allowed were wrong. The max rule committed nothing.

### 7.3 A guaranteed error among commits of 10% is reachable only for the strongest decoder

The marginal rules let a weak decoder commit text that is often wrong: 21% of the guaranteed rule's commits on our decoder, and 38–44% on sEMG cross-session (Table 2). The selective controllers (Section 5.4) bound the error among commits instead (Table 3). With a target of

**Table 1: Safety and speed on the fixed test sets.** Trust, plug-in: fitted on all fit data, no guarantee. Guaranteed rule: fit-A pipeline, clean calibration, LTT with the max rule (B2T '25: deployment splits A and B, $n_{\text{cal}} = 401$; sEMG: 143). EAC: error among commits (counts if fewer than 30 commits). Silent: uncorrected wrong final texts per item. 95% cluster-bootstrap CIs over days (B2T '25), participants (sEMG) or sentences × participants (MEG), conditional on the fitted pipeline and threshold. Zero rates: one-sided 95% upper bounds 0.0045 (816 items), 0.0054 (551) and 0.0047 (637). All policies, including pure CRC and the plain threshold: Table 9.

| Device | Policy | eWPM [CI] | Wrong-commit [CI] | Commit | EAC | Silent |
|---|---|---|---|---|---|---|
| B2T '25, our decoder | always commit | 11.45 [10.70, 12.31] | 0.610 [0.568, 0.647] | 1.00 | 0.61 | 0.044 |
| | always offer | 11.37 [10.67, 12.17] | 0 | 0 | — | 0 |
| | trust, plug-in | 11.55 [10.83, 12.34] | 0.059 [0.045, 0.070] | 0.23 | 0.25 (48/189) | 0.003 |
| | **guaranteed rule, A** | 11.48 [10.79, 12.29] | **0.025** [0.015, 0.035] | 0.10 | 0.24 (20/82) | 0.001 |
| | **guaranteed rule, B** | 11.54 [10.83, 12.34] | **0.052** [0.043, 0.059] | 0.19 | 0.27 (42/157) | 0.003 |
| B2T '25, published | always commit | 19.61 [17.81, 21.32] | 0.194 [0.146, 0.253] | 1.00 | 0.19 | 0.025 |
| | always offer | 18.11 [16.59, 19.57] | 0 | 0 | — | 0 |
| | trust gate, plug-in | 18.77 [17.13, 20.35] | 0.044 [0.027, 0.068] | 0.47 | 0.09 (36/386) | 0.004 |
| | **guaranteed rule, A** | 18.65 [17.04, 20.20] | **0.029** [0.013, 0.050] | 0.40 | 0.07 (24/329) | 0.001 |
| | **guaranteed rule, B** | 18.68 [17.06, 20.23] | **0.031** [0.014, 0.052] | 0.42 | 0.07 (25/339) | 0.003 |
| B2T '25 | gaze / gaze + completion | 4.08 [4.05, 4.12] / 5.84 | 0 | — | — | 0 |
| sEMG session-cal. | always commit | 7.81 [6.66, 9.04] | 0.541 [0.463, 0.621] | 1.00 | 0.54 | 0.060 |
| | always offer | 8.01 [6.92, 9.17] | 0 | 0 | — | 0 |
| | trust, plug-in | 8.02 [6.93, 9.19] | 0.004 [0, 0.009] | 0.04 | 2/21 | 0 |
| | **guaranteed rule** | 7.98 [6.95, 9.11] | **0.004** [0, 0.009] | 0.04 | 2/21 | 0 |
| | trust, 256 non-templated | 6.97 [5.94, 8.09] | 0.004 (1/256) | 0.01 | 1/2 | 0 |
| sEMG cross-session | always commit | 4.10 [3.56, 4.73] | 0.882 [0.831, 0.928] | 1.00 | 0.88 | 0.149 |
| | trust (= guaranteed rule) | 4.68 [4.19, 5.25] | 0.004 [0, 0.009] | 0.02 | 2/10 | 0 |
| | trust, 256 non-templated | 4.08 [3.77, 4.42] | 0 | 0 | — | 0 |
| sEMG cross-subject | always commit | 3.53 [3.19, 3.90] | 0.962 [0.942, 0.981] | 1.00 | 0.96 | 0.167 |
| | trust (= always offer) | 4.08 [3.82, 4.37] | 0 | 0 | — | 0 |
| | trust, 256 non-templated | 3.69 [3.50, 3.92] | 0 | 0 | — | 0 |
| sEMG | gaze / gaze + completion | 4.27 [4.22, 4.32] / 6.19 | 0 | — | — | 0 |
| MEG (asynchronous) | always commit | 2.58 [2.31, 2.88] | 0.997 [0.987, 1.000] | 1.00 | 1.00 | 0.235 |
| | trust (= always offer) | 3.20 [3.02, 3.44] | 0 | 0 | — | 0 |
| MEG | gaze / gaze + completion | 3.15 [3.03, 3.31] / 4.10 | 0 | — | — | 0 |

$\beta = 0.10$, only the published gate committed: SGR committed 11% of sentences on average, with an error among commits of 0.069 and a marginal wrong-commit rate of 0.008, but it committed nothing in 64% of deployments, and never exceeded $\beta$. At $\beta = 0.20$, SGR committed 62% of published sentences, but its marginal wrong-commit rate (0.088) exceeded $\alpha$; the joint rule kept both targets and equals the CRC max rule (42%, error among commits 0.085). Our decoder committed nothing at $\beta \leq 0.10$ and 0.3% of trials at $\beta = 0.20$; sEMG and MEG committed nothing at any $\beta$, so the layer equals always offering there. On the fixed splits, no selective controller committed at $\beta \leq 0.10$. The limit is the calibration size: exact binomial bounds on 10–80 calibration commits are wide (Figure 3c shows the trade-off on the test sets).

### 7.4 Calibrating on reused data raises the risk on one decoder; no within-day drift is detectable

CRC needs calibration items that are exchangeable with new items for a fixed score function. Calibrating on the data that also fitted the temperatures and the policy breaks this. In the repeated deployments, this reuse raised the mean pure-CRC test rate on our decoder from 0.046 to 0.055 (paired difference +0.009, 5th–95th percentile −0.020 to +0.040; higher in 35 of 49 deployments with a difference; Wilcoxon $p = 0.003$), above $\alpha$, and the share of deployments above $\alpha$ from 0.34 to 0.58 (Table 2). On the published gate it changed nothing (+0.001, $p = 0.72$), on sEMG session-calibrated −0.002 ($p = 0.71$), and on sEMG cross-session +0.007 (Wilcoxon $p = 0.036$, sign test $p = 0.47$). The effect is real for one decoder but small next to the spread of one deployment: a single deployment detects a shift of 0.035 (our decoder), 0.038 (published) or 0.040–0.057 (sEMG) with 80% power.

On the fixed deployment splits (Table 16 in Appendix M), clean CRC gave 0.0245 and 0.0600

**Table 2: Wrong-commit control over 50 repeated deployments.** Each deployment re-draws fit-A, clean calibration and test data (Section 6) and refits the whole pipeline. Wrong-commit: mean test rate (5th–95th percentile over deployments); $> \alpha$: share of deployments with a test rate above $\alpha = 0.05$ (LTT allows $\delta = 0.1$); Commit: mean commit rate; EAC: error among commits pooled over deployments. Reused: pure CRC calibrated on the data that also fitted the pipeline. sEMG cross-subject never committed under any rule. The spread is split-to-split variation in one data set, not a CI for new data.

| | B2T '25, our decoder | | | | | B2T '25, published gate | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Rule** | **Wrong-commit** | **$> \alpha$** | **Commit** | **EAC** | **eWPM** | **Wrong-commit** | **$> \alpha$** | **Commit** | **EAC** | **eWPM** |
| plug-in (fit-A) | 0.044 (0.021–0.072) | 0.34 | 0.19 | 0.24 | 11.08 | 0.039 (0.021–0.060) | 0.22 | 0.43 | 0.09 | 18.54 |
| CRC, pure | 0.046 (0.026–0.068) | 0.34 | 0.19 | 0.24 | 11.09 | 0.049 (0.029–0.075) | 0.42 | 0.48 | 0.10 | 18.60 |
| CRC, max rule | 0.037 (0.021–0.059) | 0.14 | 0.17 | 0.22 | 11.08 | 0.036 (0.021–0.055) | 0.14 | 0.42 | 0.09 | 18.52 |
| LTT, pure | 0.034 (0.017–0.056) | 0.16 | 0.16 | 0.22 | 11.07 | 0.034 (0.021–0.051) | 0.06 | 0.41 | 0.08 | 18.51 |
| **LTT, max rule** | **0.030** (0.016–0.052) | **0.08** | 0.15 | 0.21 | 11.07 | **0.031** (0.020–0.047) | **0.02** | 0.39 | 0.08 | 18.49 |
| CRC, pure, reused | 0.055 (0.037–0.085) | 0.58 | 0.21 | 0.26 | 11.10 | 0.050 (0.032–0.073) | 0.48 | 0.48 | 0.10 | 18.61 |
| | **sEMG session-calibrated** | | | | | **sEMG cross-session** | | | | |
| plug-in (fit-A) | 0.015 (0.004–0.033) | 0 | 0.12 | 0.13 | 8.01 | 0.008 (0–0.033) | 0.04 | 0.02 | 0.44 | 4.68 |
| CRC, pure | 0.029 (0.011–0.061) | 0.06 | 0.19 | 0.15 | 8.04 | 0.025 (0–0.063) | 0.14 | 0.04 | 0.58 | 4.67 |
| CRC, max rule | 0.013 (0.004–0.024) | 0 | 0.11 | 0.12 | 8.01 | 0.007 (0–0.027) | 0 | 0.02 | 0.41 | 4.68 |
| **LTT, max rule** | **0.011** (0.004–0.019) | **0** | 0.09 | 0.12 | 8.00 | **0.006** (0–0.023) | **0** | 0.02 | 0.38 | 4.68 |

**Table 3: Controlling the error among commits** (target $\beta$, $\delta = 0.1$; 50 repeated deployments; test set). Commit: mean (5th–95th percentile); None: share of deployments without a commit; EAC: pooled error among commits; $> \beta$: share of deployments with EAC above $\beta$; Marginal: mean wrong-commit rate. Joint: $\max(\lambda_{\text{SGR}}, \lambda_{\text{CRC max}})$. Our decoder at $\beta \leq 0.10$, sEMG (all conditions) and MEG (sentence-grouped splits) at every $\beta$: no commits.

| **Decoder** | **$\beta$** | **Rule** | **Commit** | **None** | **EAC** | **$> \beta$** | **Marginal** | **eWPM** |
|---|---|---|---|---|---|---|---|---|
| B2T '25, published | 0.10 | SGR | 0.112 (0–0.376) | 0.64 | 0.069 | 0 | 0.008 | 18.12 |
| | 0.10 | LTT-sel | 0.072 (0–0.335) | 0.76 | 0.059 | 0 | 0.004 | 18.07 |
| | 0.10 | joint | 0.111 (0–0.367) | 0.64 | 0.068 | 0 | 0.008 | 18.12 |
| | 0.20 | SGR | 0.617 (0.455–0.850) | 0 | 0.142 | 0 | 0.088 | 18.79 |
| | 0.20 | LTT-sel | 0.564 (0.361–0.730) | 0 | 0.131 | 0 | 0.074 | 18.72 |
| | 0.20 | joint | 0.418 (0.356–0.494) | 0 | 0.085 | 0 | 0.036 | 18.52 |
| | — | no commits | 0 | 1 | — | — | 0 | 17.98 |
| B2T '25, our decoder | 0.20 | SGR | 0.003 (0–0) | 0.96 | 0.208 | 0.02 | 0.001 | 10.95 |
| | — | no commits | 0 | 1 | — | — | 0 | 10.95 |

on our decoder, 0.0478 and 0.0490 on the published gate, and 0.0563 and 0.0399 on sEMG cross-session and session-calibrated, against 0.0625, 0.0490, 0.0635 and 0.0417 when calibrated on reused fit data with the full-fit pipeline. The two clean values above $\alpha$ were within the range of exchangeable splits ($p = 0.21$ and 0.30). In split A, the test rate was above the exchangeable reference ($p = 0.012$; day-stratified reference $p = 0.014$) but below $\alpha$, where CRC was not binding; after Holm correction over the six tests, no test was significant (split A: 0.072). These tests have low power: the 5th–95th percentile range of the reference is 0.029–0.071 on Brain-to-Text '25. We therefore claim neither within-day drift nor that data reuse caused the earlier value of 0.0625, which is within the clean protocol's split-to-split range; drift, reuse and split variation cannot be fully separated on these data. Reusing fit data for calibration is invalid in principle, and all guaranteed rows use clean calibration. The online controllers kept the long-run rate at 0.0245–0.0527 (Brain-to-Text '25) and 0.0381–0.0526 (sEMG), at eWPM within 0.06 of offline CRC, but rolling 50-item windows reached 0.06–0.18, and when only noticed errors were fed back the long-run rate reached 0.027–0.058 (Appendix M).

### 7.5 Speed rankings depend on the user model; wrong-commit rates do not

Figure 2 refits every policy under eight user-model variants (plug-in layer). The layer was at least as fast as always offering in every variant and condition (on Brain-to-Text '25 all intervals above

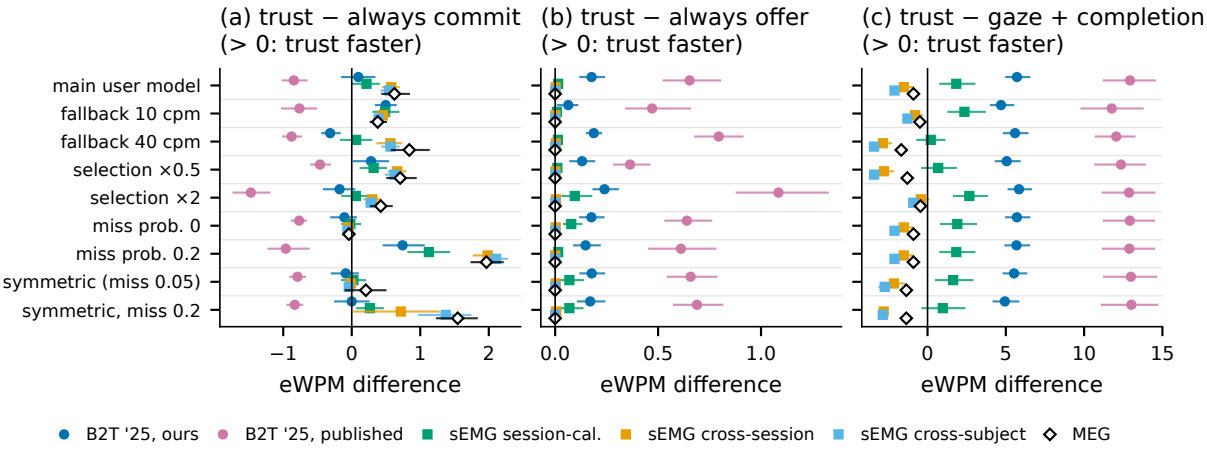


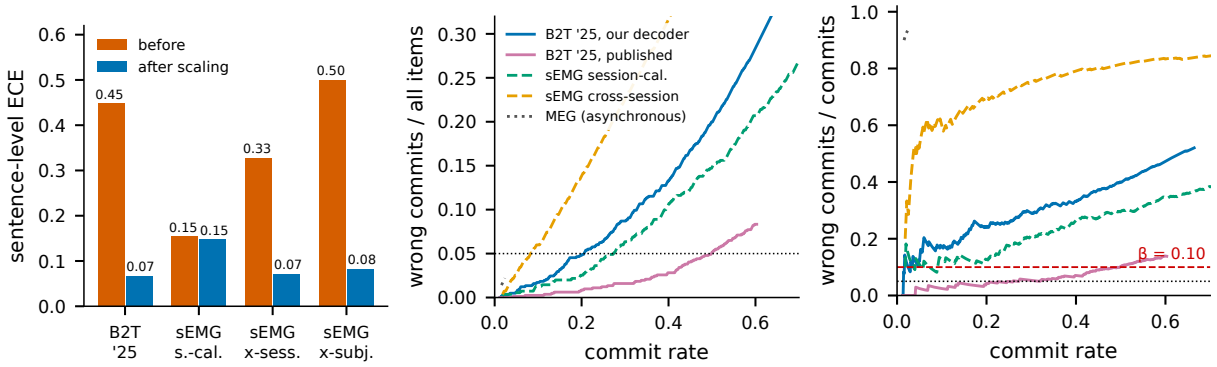

**Figure 2: User-model sensitivity** (plug-in layer). Paired eWPM difference between the trust layer and (a) always committing, (b) always offering and (c) gaze typing with word completion, all policies refitted under each variant (rows); values above 0 mean that the layer is faster. 95% cluster-bootstrap CIs at the primary unit ($B = 2000$).

0). The comparison with always committing depends on the user model: on our intracortical decoder, always committing was at least as fast in 5 of 8 variants (fallback 40 characters per minute, selection ×2, miss probability 0, both symmetric-offer variants), and on the published decoder it was the fastest policy in all 9, by 0.46–1.47 eWPM. On sEMG and MEG, the gain over always committing came from offers: it disappeared with miss probability 0 and under symmetric offers with miss probability 0.05, but remained under symmetric offers with miss probability 0.2 (MEG [+1.25, +1.82]; sEMG cross-subject [+0.98, +1.73]; session-calibrated [+0.08, +0.45]). Gaze typing with word completion had a higher point value than the layer in every variant on MEG and on the two weaker sEMG decoders, and the MEG comparison with plain gaze typing flipped either way (fallback 40 characters per minute [−1.35, −0.91]; 10 characters per minute [+0.21, +0.44]). In contrast, the wrong-commit rate of the refitted layer was unchanged in 43 of the 48 non-main condition–variant pairs and at most 0.059 in all (changes: sEMG session-calibrated in four variants, 0.004 to 0.020; cross-session with miss probability 0.2, to 0). Silent errors rose under symmetric offers (up to 0.17), except on MEG. Policies fitted under the main user lost at most 0.2 eWPM against refitting, except with symmetric offers and miss probability 0.2 on the weak decoders (MEG 1.18 vs 2.75).

### 7.6 What makes the layer safe: sentence calibration and a threshold, not the guard

**Calibration.** Sentence ECE fell from 0.448 to 0.067 on our intracortical decoder (frame temperature 3.78; the ECE of per-frame phoneme peaks rose from 0.090 to 0.130), from 0.102 to 0.058 on the published gate, and from 0.154, 0.328 and 0.500 to 0.148, 0.072 and 0.083 on sEMG (Figure 3a); on MEG and sEMG cross-subject the sentence temperature reached the search bound and flattens the confidence. The calibrated top-1 probability predicted a wrong top sentence with AUROC 0.797 (intracortical) and 0.83, 0.76 and 0.67 (sEMG). We did not compute CIs for ECE and AUROC.

**Ablation** (Table 10, Appendix H). (1) The *sentence* temperature keeps wrong commits low: without it, the rate rose on 4 of 6 conditions (our intracortical decoder 0.076, paired [+0.001, +0.036]; sEMG 0.040, 0.018, 0.044). (2) The *frame* temperature changes decoding, because it is applied before an LM search whose weight was tuned on raw streams. On sEMG cross-subject, re-tuning the LM weight on the calibrated streams gave 4.84 [4.28, 5.51] eWPM




**(a)** Sentence-level ECE. **(b)** Marginal risk. **(c)** Error among commits.

**Figure 3: Calibration and the commit trade-off** (full-fit pipeline, test sets). (a) Sentence-level expected calibration error before and after temperature scaling. (b) For each commit threshold on the calibrated score: share of test items committed (x) and committed wrongly (y); dotted line: $\alpha = 0.05$. (c) The same thresholds: error among commits; dashed line: $\beta = 0.10$. Items blocked by the guard or other policy rules are never committed. Curves start at 10 commits. sEMG cross-subject is absent because no candidate passed the guard; the MEG curve has fewer than 10 commits above most thresholds.

at top-1 CER 0.360 (instead of 0.479); the difference to gaze typing [−0.00, +1.25] includes 0, and it stays below gaze typing with word completion. (3) The guard did not help on real data: without it, the wrong-commit rate fell from 0.059 to 0.036 on our intracortical decoder (paired [−0.035, −0.011]; lower on 9 of 10 days with a difference, sign test $p = 0.021$; eWPM [−0.10, +0.04]) and from 0.044 to 0.025 on the published gate ([−0.034, −0.007]; eWPM [−0.20, −0.10]), and changed nothing on sEMG and MEG. Its AUROC for a wrong top sentence (0.702) was below that of $\hat{p}_{\text{top1}}$. Conformal sets and OFFER_WORDS changed little. We report the configuration fixed before evaluation (with guard) because choosing it from a test-set ablation would bias the reported rates downwards, and because the plain threshold was not run on the control streams, where the guard separated signal-destroying streams from real ones (Section 7.7). The frozen configuration for a confirmatory test uses the plain threshold (Section 8).

**The commit trade-off.** As the threshold varies (Figures 3b and 3c), the published decoder can commit about half of its sentences at the 0.05 marginal level and our decoder about a fifth. The error among commits of our decoder falls to 0.1–0.2 only below about 10% commits and rises to about 0.5 at 60%, while that of the published decoder stays near or below 0.15; on sEMG cross-session, most commits are wrong at all but the lowest rates.

### 7.7 Controls: signal-destroying streams are not committed; intent mismatch is not detectable

Always commit wrote a sentence for every control stream (on MEG, 85–100% fluent Spanish). The trust layer committed none of the signal-destroying controls on Brain-to-Text '25 (0 of 300 trials per control; one-sided 95% upper bound 0.0099) or sEMG (0 of 200 sentences per condition; bound 0.015) (Figure 7a, b in Appendix O), an informative result because the same layer committed 22% and 2.5–4% of the real inputs in the same audit. On Brain-to-Text '25, the guard risk separated frame-shuffled and uniform streams from real ones with AUROC 0.981 and 0.998. On MEG, the layer committed nothing on real input either, so its zero commits on controls carry no information; the guard risk separated shuffled MEG, zero input and uniform streams (AUROC 0.92, 1.00, 1.00), but not frame-shuffled streams (0.47) (Figure 7c). Given the real stream of another sentence, the layer committed at the matched rate on our intracortical decoder (22%) and on sEMG: nothing in the stream reveals the intent. The published gate

committed none of these, but only because its text came from the matched trial.

# 8 Discussion

**The layer is a safety mechanism, tested non-trivially on one person.** All non-trivial evidence for the commit decision comes from one intracortical participant in a copy task; sEMG sentences were composed from isolated presses, and MEG was a negative control. On that participant, a stream-only rule with a high-probability guarantee kept the wrong-commit rate at or below 5% in 92% (our decoder) and 98% (published decoder) of repeated deployments, at a mean of 3%, and it stayed faster than always offering in every deployment, by 0.11 and 0.51 eWPM. Whether it beat always committing depended on unmeasured user costs, and on weaker decoders a gaze keyboard with simple word completion was faster still, so we make no general speed claim. On sEMG and MEG the layer reduced to always offering, the correct behaviour for decoders whose top sentence was rarely right; it cannot create information that the decoder lacks.

**Which guarantee to deploy.** CRC met its target as a mean over deployments, exactly as it promises, but a third or more of single deployments exceeded $\alpha$; so did the plug-in rule. A user lives in one deployment, so the per-deployment (high-probability) guarantee of LTT is the relevant one, and the max rule costs almost nothing (0.01–0.04 eWPM against CRC with the max rule). Each guarantee should be stated with its assumptions, its target, its unit of exchangeability and the measured silent-error rate (Appendix P). On MEG the unit is the sentence, and no guarantee for new sentences is possible with 17 test sentences.

**Who sets $\alpha$, and for which error.** The marginal wrong-commit rate is the budget of harm per sentence attempted; the error among commits is what a user sees in committed text. With current decoders the two diverge: one in five commits of the guaranteed rule on our decoder was wrong, and a user who sees this will learn to check every commit, which turns commits into offers. A guaranteed error among commits of 10% was reachable only for the published decoder, and only in about a third of deployments with 400 calibration trials. With a few hundred calibration sentences, a certified "one wrong committed sentence in ten" is possible only for the strongest decoder; the joint rule (marginal $\leq \alpha$ and selective $\leq \beta$) is the one to recommend. The targets should come from users. Potential BCI users with ALS or spinal cord injury asked for at least 90% accuracy [12, 13], which suggests $\beta \leq 0.10$ rather than a marginal target alone. The choice should be made by the user with their clinician and communication partners, explained in these terms ("at most one in ten sentences that the system sends by itself is wrong"), and it may differ by context: a casual conversation tolerates more than a medical decision or a message to a stranger. AAC users care about text that is sent in their name [16, 17], and our exact-match loss treats a harmless near miss ("who you never expect to when") like a meaning-changing error ("I want to stop" vs "I want to start"). A loss weighted by semantic severity, scored by users, is the natural next target.

**Calibration data must play no other role.** Calibrating on data that had also fitted the pipeline raised the mean CRC wrong-commit rate on our decoder above $\alpha$, but not on the published gate or sEMG, and the effect was a quarter of one deployment's spread; drift, reuse and split variation cannot be separated on one fixed split. In data-scarce BCI pipelines this mistake is easy. Papers should state which data fitted each stage and repeat the fit/calibration assignment before they claim a shift or a guarantee.

**Strong decoders still need a gate.** The published Brain-to-Text '25 decoder has a WER of 0.043, yet 19% of its sentences were wrong. Gating its text by agreement with the neural

evidence kept 96% of the always-commit eWPM under the main simulated user and reduced wrong commits to 4.4% (plug-in) and 3.1% (guaranteed rule, mean over deployments), using only the stream, the published text and a light reference decoder. This is what a vendor-neutral stream interface enables; on sEMG, in-session decoder data gave 71% higher eWPM than another day's data (8.02 vs 4.68), so adaptation, not the layer, makes a weak decoder fast. The trust layer took a median of 0.18–0.30 ms per decision on one CPU thread (95th percentile at most 1.1 ms), against 45–232 ms for the beam-search decoders, so it adds no meaningful latency.

**A frozen configuration for a confirmatory test.** Every revision of this paper looked at the same Brain-to-Text '25 test half, so all results here are exploratory. For a confirmatory test on data we have not used (a second participant, or held-out sessions), we fix the configuration now: frame and sentence temperatures, plain calibrated threshold without the guard, policy grid of Table 8 fitted on fit-A, LTT with the max rule on a disjoint calibration set, $\alpha = 0.05$, $\delta = 0.1$, and the joint selective constraint with $\beta = 0.10$; primary outcomes are the wrong-commit rate and the error among commits, with their share of deployments above target. This configuration is tagged `paper1-v3-frozen-config` in the repository.

## 9 Limitations

1. **One implant participant** in a copy task, with online logits we cannot re-run, is the only non-trivial test of the commit decision; our intervals cover days and deployments of this person, not people. A second public intracortical data set was not analysed.
2. **Composed sEMG sentences**: isolated presses with oracle timing, recurring presses, and test sentences from the LM's corpus. 295 of 551 test sentences have a near variant in the LM training text. Rebuilding the LM without them changed trust eWPM by at most $-0.04$, but excluding them from the test set lowered trust eWPM from 8.02, 4.68 and 4.08 to 6.97, 4.08 and 3.69, and made gaze typing faster than the layer on the cross-subject decoder (Appendix I). **MEG is a negative control**: 17 unique test sentences (too few for a sentence-level guarantee) and a decoder that uses key times as training labels.
3. **No live users.** Only two user costs are measured, in one gaze-typing user in a non-BCI setting; the miss probability is unmeasured; the MEG and sEMG populations do not match; speed rankings change across plausible variants; the word-completion baseline is a context-free unigram, weaker than clinical AAC. The loss is exact match, which ignores the severity of an error. A user study needs ethics approval.
4. **Repeated splits re-use one data set.** The 50 deployments re-draw fit, calibration and test data from the same trials (Brain-to-Text '25) or the same fit sentences (sEMG); their spread is not a CI for new sessions or people. Decoder seeds were repeated only for MEG. Trial order is file order (no time stamps), and the sEMG online order is not a time sequence.
5. **Scope of the guarantees.** CRC bounds the mean over deployments, LTT the population risk with probability $1-\delta$, and the online methods a long-run average; all count only observed errors. A guarantee on the error among commits was reachable only for the published decoder. Stream-only gating cannot detect intent mismatch, and the audit log is tamper-evident, not tamper-proof.

## 10 Conclusion

A trust layer that reads only a per-frame output stream can decide whether a neural text-entry decoder commits, offers or abstains, with a high-probability bound on the wrong-commit rate. On one intracortical participant, Learn-then-Test with the max rule kept wrong commits at

or below 5% in 92–98% of repeated deployments at little speed cost relative to always offering, while about one in five (our decoder) or one in thirteen (published decoder) committed sentences was still wrong; conformal risk control and the plug-in rule met the target only on average. A guaranteed error among commits of 10% was reachable only for the strongest decoder, and on composed sEMG sentences and a weak MEG decoder the layer almost never or never committed. Calibration data must play no other role. Next steps are a confirmatory test of the frozen configuration on new data, a user study that measures which error rates users accept, and the public benchmark that the release starts.

## 11 Ethics, data and code availability

This study uses only public, de-identified datasets; no new data were collected and no experiment with human participants was run, so no new ethics approval was required. The intracortical data come from a participant in the BrainGate2 pilot clinical trial [2]; the released copy-task file is CC0, we use only its logits and cue sentences, and we did not attempt re-identification. Quoted sentences are public copy-task cues, not the participant's own words. **[TODO: author decision: final ethics wording and the venue's policy on secondary use of public clinical-trial data.]** No raw recordings or participant data are redistributed; Table 11 lists all datasets and licences. The NeuralTextStream specification, conformance suite, reference implementation, benchmark scorer and all experiment scripts are at `https://github.com/zhangzibo/fidetype` (Apache-2.0), without weights trained on SpanishBCBL or copied Brain2Qwerty code. Every table and figure is generated from committed result files by committed scripts; the experiment-level result files themselves need the public raw datasets and a local decode cache, which the repository's experiment folders describe with their exact commands. **[TODO: archive the paper commit with a DOI (e.g. Zenodo).]** Decoders were trained on one AMD RX 7800 XT GPU. The trust layer runs on a CPU: on one thread of an AMD Ryzen 7 9800X3D, it took a median of 0.18–0.30 ms per decision (95th percentile 0.49–1.06 ms), measured while other jobs ran, against 45–232 ms for the beam-search decoders.

## Acknowledgements

**[TODO: funding.]** AI coding and writing assistants were used under human review; all numbers come from committed scripts and result files. **[TODO: author: final AI-assistance wording.]**

## References

[1] Francis R. Willett, Erin M. Kunz, Chaofei Fan, Donald T. Avansino, Guy H. Wilson, Eun Young Choi, Foram Kamdar, Matthew F. Glasser, Leigh R. Hochberg, Shaul Druckmann, Krishna V. Shenoy, and Jaimie M. Henderson. A high-performance speech neuroprosthesis. *Nature*, 2023. doi: 10.1038/s41586-023-06377-x.

[2] Nicholas S. Card, Maitreyee Wairagkar, Carrina Iacobacci, Xianda Hou, Tyler Singer-Clark, Francis R. Willett, Erin M. Kunz, Chaofei Fan, Maryam Vahdati Nia, Darrel R. Deo, Aparna Srinivasan, Eun Young Choi, Matthew F. Glasser, Leigh R. Hochberg, Jaimie M. Henderson, Kiarash Shahlaie, Sergey D. Stavisky, and David M. Brandman. An accurate and rapidly calibrating speech neuroprosthesis. *New England Journal of Medicine*, 391(7): 609–618, 2024. doi: 10.1056/NEJMoa2314132.

[3] Sean L. Metzger, Kaylo T. Littlejohn, Alexander B. Silva, David A. Moses, Margaret P. Seaton, Ran Wang, Maximilian E. Dougherty, Jessie R. Liu, Peter Wu, Michael A. Berger,

Inga Zhuravleva, Adelyn Tu-Chan, Karunesh Ganguly, Gopala K. Anumanchipalli, and Edward F. Chang. A high-performance neuroprosthesis for speech decoding and avatar control. *Nature*, 620(7976):1037–1046, 2023. doi: 10.1038/s41586-023-06443-4.

[4] Francis R. Willett, Donald T. Avansino, Leigh R. Hochberg, Jaimie M. Henderson, and Krishna V. Shenoy. High-performance brain-to-text communication via handwriting. *Nature*, 593(7858):249–254, 2021. doi: 10.1038/s41586-021-03506-2.

[5] Justin J. Jude, Hadar Levi-Aharoni, Alexander J. Acosta, Shane B. Allcroft, Claire Nicolas, Bayardo E. Lacayo, Nicholas S. Card, Maitreyee Wairagkar, Alisa D. Levin, David M. Brandman, Sergey D. Stavisky, Francis R. Willett, Ziv M. Williams, John D. Simeral, Leigh R. Hochberg, and Daniel B. Rubin. Restoring rapid natural bimanual typing with a neuroprosthesis after paralysis. *Nature Neuroscience*, 29(10):2613–2622, 2026. doi: 10.1038/s41593-026-02218-y.

[6] Jarod Lévy, Mingfang Zhang, Svetlana Pinet, Jérémy Rapin, Hubert Banville, Stéphane d'Ascoli, and Jean-Rémi King. Noninvasive decoding of typed sentences from human brain activity. *Nature Neuroscience*, 29:2438–2444, 2026. doi: 10.1038/s41593-026-02303-2.

[7] Mingfang Zhang, Jarod Lévy, Cedric Rommel, Jérémy Rapin, Corentin Bel, Julie Bonnaire, Daniel Nieto, Pierre Bourdillon, Svetlana Pinet, Stéphane d'Ascoli, Thomas Moreau, and Jean-Rémi King. Accurate decoding of natural sentences from non-invasive brain recordings. Meta preprint, https://facebookresearch.github.io/brain2qwerty/assets/brain2qwerty_v2.pdf, 2026. 29 June 2026.

[8] Viswanath Sivakumar, Jeffrey Seely, Alan Du, Sean R. Bittner, Adam Berenzweig, Anuoluwapo Bolarinwa, Alexandre Gramfort, and Michael I. Mandel. emg2qwerty: A large dataset with baselines for touch typing using surface electromyography, 2024. arXiv:2410.20081.

[9] Jonathan Eby, Moshe Beutel, David Koivisto, Idan Achituve, Ethan Fetaya, and José Zariffa. Electromyographic typing gesture classification dataset for neurotechnological human-machine interfaces. *Scientific Data*, 12:440, 2025. doi: 10.1038/s41597-025-04763-w.

[10] Patrick Kaifosh, Thomas R. Reardon, and CTRL-labs at Reality Labs. A generic non-invasive neuromotor interface for human-computer interaction. *Nature*, 645(8081):702–711, 2025. doi: 10.1038/s41586-025-09255-w.

[11] Allison Koenecke, Anna Seo Gyeong Choi, Katelyn X. Mei, Hilke Schellmann, and Mona Sloane. Careless whisper: Speech-to-text hallucination harms. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2024. doi: 10.1145/3630106.3658996.

[12] Jane E. Huggins, Patricia A. Wren, and Kirsten L. Gruis. What would brain-computer interface users want? Opinions and priorities of potential users with amyotrophic lateral sclerosis. *Amyotrophic Lateral Sclerosis*, 12(5):318–324, 2011. doi: 10.3109/17482968.2011.572978.

[13] Jane E. Huggins, Aisha A. Moinuddin, Anthony E. Chiodo, and Patricia A. Wren. What would brain-computer interface users want: opinions and priorities of potential users with spinal cord injury. *Archives of Physical Medicine and Rehabilitation*, 96(3 Suppl):S38–S45.e5, 2015. doi: 10.1016/j.apmr.2014.05.028.

[14] Christine H. Blabe, Vikash Gilja, Cindy A. Chestek, Krishna V. Shenoy, Kim D. Anderson, and Jaimie M. Henderson. Assessment of brain–machine interfaces from the perspective of

people with paralysis. *Journal of Neural Engineering*, 12(4):043002, 2015. doi: 10.1088/1741-2560/12/4/043002.

[15] Andrea Kübler, Elisa M. Holz, Angela Riccio, Claudia Zickler, Tobias Kaufmann, Sonja C. Kleih, Pit Staiger-Sälzer, Lorenzo Desideri, Evert-Jan Hoogerwerf, and Donatella Mattia. The user-centered design as novel perspective for evaluating the usability of BCI-controlled applications. *PLoS ONE*, 9(12):e112392, 2014. doi: 10.1371/journal.pone.0112392.

[16] Stephanie Valencia, Richard Cave, Krystal Kallarackal, Katie Seaver, Michael Terry, and Shaun K. Kane. "The less I type, the better": How AI language models can enhance or impede communication for AAC users. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*, pages 1–14. ACM, 2023. doi: 10.1145/3544548.3581560.

[17] Shaun K. Kane, Meredith Ringel Morris, Ann Paradiso, and Jon Campbell. "At times avuncular and cantankerous, with the reflexes of a mongoose": Understanding self-expression through augmentative and alternative communication devices. In *Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing (CSCW '17)*, pages 1166–1179. ACM, 2017. doi: 10.1145/2998181.2998284.

[18] Maitreyee Wairagkar, Nicholas S. Card, Tyler Singer-Clark, Xianda Hou, Carrina Iacobacci, Lee M. Miller, Leigh R. Hochberg, David M. Brandman, and Sergey D. Stavisky. An instantaneous voice-synthesis neuroprosthesis. *Nature*, 644(8075):145–152, 2025. doi: 10.1038/s41586-025-09127-3.

[19] Erin M. Kunz, Benyamin Abramovich Krasa, Foram Kamdar, Donald T. Avansino, Nick Hahn, Seonghyun Yoon, Akansha Singh, Samuel R. Nason-Tomaszewski, Nicholas S. Card, Justin J. Jude, Brandon G. Jacques, Payton H. Bechefsky, Carrina Iacobacci, Leigh R. Hochberg, Daniel B. Rubin, Ziv M. Williams, David M. Brandman, Sergey D. Stavisky, Nicholas AuYong, Chethan Pandarinath, Shaul Druckmann, Jaimie M. Henderson, and Francis R. Willett. Inner speech in motor cortex and implications for speech neuroprostheses. *Cell*, 188(17):4658–4673.e17, 2025. doi: 10.1016/j.cell.2025.06.015.

[20] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017. arXiv:1706.04599.

[21] Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/hash/4a8423d5e91fda00bb7e46540e2b0cf1-Abstract.html. arXiv:1705.08500.

[22] Stephen Bates, Anastasios Angelopoulos, Lihua Lei, Jitendra Malik, and Michael I. Jordan. Distribution-free, risk-controlling prediction sets. *Journal of the ACM*, 68(6):43:1–43:34, 2021. doi: 10.1145/3478535.

[23] Anastasios N. Angelopoulos, Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster. Conformal risk control. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2208.02814.

[24] Anastasios N. Angelopoulos, Stephen Bates, Emmanuel J. Candès, Michael I. Jordan, and Lihua Lei. Learn then test: Calibrating predictive algorithms to achieve risk control, 2021. arXiv:2110.01052.

[25] Soowon Kim, Byung-Kwan Ko, and Seo-Hyun Lee. Confidence-aware neural decoding of overt speech from EEG: Toward robust brain–computer interfaces, 2025. arXiv:2511.07890.

[26] Jingya Huang, Sowmya Manojna Narasimha, Aashish N. Patel, Gal Mishne, and Vikash Gilja. Prototyping co-control brain–computer interfaces through brain-to-text. In *Data on the Brain & Mind Workshop (Findings track), NeurIPS 2025*, 2025. URL https://openreview.net/forum?id=vECJkkSzEX.

[27] Amit Damri and Bracha Laufer-Goldshtein. Confident and adaptive generative speech recognition via risk control. In *International Conference on Learning Representations (ICLR)*, 2026. URL https://openreview.net/forum?id=ck5T7QeiDh.

[28] Shanqing Cai, Subhashini Venugopalan, Katie Seaver, Xiang Xiao, Katrin Tomanek, Sri Jalasutram, Meredith Ringel Morris, Shaun Kane, Ajit Narayanan, Robert L. MacDonald, Emily Kornman, Daniel Vance, Blair Casey, Steve M. Gleason, Philip Q. Nelson, and Michael P. Brenner. Using large language models to accelerate communication for eye gaze typing users with ALS. *Nature Communications*, 2024. doi: 10.1038/s41467-024-53873-3.

[29] Alexandre Défossez, Charlotte Caucheteux, Jérémy Rapin, Ori Kabeli, and Jean-Rémi King. Decoding speech perception from non-invasive brain recordings. *Nature Machine Intelligence*, 2023. doi: 10.1038/s42256-023-00714-5.

[30] Dulhan Jayalath, Gilad Landau, and Oiwi Parker Jones. Unlocking non-invasive brain-to-text, 2025. arXiv:2505.13446.

[31] Dulhan Jayalath and Oiwi Parker Jones. MEG-XL: Data-efficient brain-to-text via long-context pre-training, 2026. arXiv:2602.02494.

[32] Sean L. Metzger, Jessie R. Liu, David A. Moses, Maximilian E. Dougherty, Margaret P. Seaton, Kaylo T. Littlejohn, Josh Chartier, Gopala K. Anumanchipalli, Adelyn Tu-Chan, Karunesh Ganguly, and Edward F. Chang. Generalizable spelling using a speech neuroprosthesis in an individual with severe limb and vocal paralysis. *Nature Communications*, 2022. doi: 10.1038/s41467-022-33611-3.

[33] Francis R. Willett, Jingyuan Li, Trung Le, Chaofei Fan, Mingfei Chen, Eli Shlizerman, et al. Brain-to-text benchmark '24: Lessons learned, 2024. arXiv:2412.17227.

[34] Nicholas S. Card et al. Data for: An accurate and rapidly calibrating speech neuroprosthesis. Dryad, version 5, 2025. CC0 1.0.

[35] Alex Graves, Santiago Fernández, Faustino Gomez, and Jürgen Schmidhuber. Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks. In *Proceedings of the 23rd International Conference on Machine Learning (ICML)*, pages 369–376, 2006. doi: 10.1145/1143844.1143891.

[36] Awni Hannun, Carl Case, Jared Casper, Bryan Catanzaro, Greg Diamos, Erich Elsen, Ryan Prenger, Sanjeev Satheesh, Shubho Sengupta, Adam Coates, and Andrew Y. Ng. Deep speech: Scaling up end-to-end speech recognition, 2014. arXiv:1412.5567.

[37] David A. Moses, Sean L. Metzger, Jessie R. Liu, Gopala K. Anumanchipalli, Joseph G. Makin, Pengfei F. Sun, Josh Chartier, Maximilian E. Dougherty, Patricia M. Liu, Gary M. Abrams, Adelyn Tu-Chan, Karunesh Ganguly, and Edward F. Chang. Neuroprosthesis for decoding speech in a paralyzed person with anarthria. *New England Journal of Medicine*, 385(3):217–227, 2021. doi: 10.1056/NEJMoa2027540.

[38] Martijn Schreuder, Johannes Höhne, Benjamin Blankertz, Stefan Haufe, Thorsten Dickhaus, and Michael Tangermann. Optimizing event-related potential based brain–computer interfaces: a systematic evaluation of dynamic stopping methods. *Journal of Neural Engineering*, 10(3):036025, 2013. doi: 10.1088/1741-2560/10/3/036025.

[39] Chandra S. Throckmorton, Kenneth A. Colwell, David B. Ryan, Eric W. Sellers, and Leslie M. Collins. Bayesian approach to dynamically controlling data collection in P300 spellers. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 21(3):508–517, 2013. doi: 10.1109/TNSRE.2013.2253125.

[40] Boyla O. Mainsah, Leslie M. Collins, Kenneth A. Colwell, Eric W. Sellers, David B. Ryan, K. Caves, and Chandra S. Throckmorton. Increasing BCI communication rates with dynamic stopping towards more practical use: an ALS study. *Journal of Neural Engineering*, 12(1):016013, 2015. doi: 10.1088/1741-2560/12/1/016013.

[41] Umut Orhan, Kenneth E. Hild, Deniz Erdogmus, Brian Roark, Barry Oken, and Melanie Fried-Oken. RSVP keyboard: An EEG based typing interface. In *2012 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 645–648, 2012. doi: 10.1109/ICASSP.2012.6287966.

[42] Barry S. Oken, Umut Orhan, Brian Roark, Deniz Erdogmus, Andrew Fowler, Aimee Mooney, Betts Peters, Meghan Miller, and Melanie B. Fried-Oken. Brain–computer interface with language model–electroencephalography fusion for locked-in syndrome. *Neurorehabilitation and Neural Repair*, 28(4):387–394, 2014. doi: 10.1177/1545968313516867.

[43] William Speier, Corey Arnold, and Nader Pouratian. Integrating language models into classifiers for BCI communication: a review. *Journal of Neural Engineering*, 13(3):031002, 2016. doi: 10.1088/1741-2560/13/3/031002.

[44] Gerwin Schalk, Jonathan R. Wolpaw, Dennis J. McFarland, and Gert Pfurtscheller. EEG-based communication: presence of an error potential. *Clinical Neurophysiology*, 111(12): 2138–2144, 2000. doi: 10.1016/S1388-2457(00)00457-0.

[45] Pierre W. Ferrez and José del R. Millán. Error-related EEG potentials generated during simulated brain–computer interaction. *IEEE Transactions on Biomedical Engineering*, 55 (3):923–929, 2008. doi: 10.1109/TBME.2007.908083.

[46] Bernardo Dal Seno, Matteo Matteucci, and Luca Mainardi. Online detection of P300 and error potentials in a BCI speller. *Computational Intelligence and Neuroscience*, 2010:307254, 2010. doi: 10.1155/2010/307254.

[47] Martin Spüler, Michael Bensch, Sonja Kleih, Wolfgang Rosenstiel, Martin Bogdan, and Andrea Kübler. Online use of error-related potentials in healthy users and people with severe motor impairment increases performance of a P300-BCI. *Clinical Neurophysiology*, 123(7):1328–1337, 2012. doi: 10.1016/j.clinph.2011.11.082.

[48] Ricardo Chavarriaga, Aleksander Sobolewski, and José del R. Millán. Errare machinale est: the use of error-related potentials in brain-machine interfaces. *Frontiers in Neuroscience*, 8:208, 2014. doi: 10.3389/fnins.2014.00208.

[49] S. G. Mason and G. E. Birch. A brain-controlled switch for asynchronous control applications. *IEEE Transactions on Biomedical Engineering*, 47(10):1297–1307, 2000. doi: 10.1109/10.871402.

[50] Haihong Zhang, Cuntai Guan, and Chuanchu Wang. Asynchronous P300-based brain–computer interfaces: A computational approach with statistical models. *IEEE Transactions on Biomedical Engineering*, 55(6):1754–1763, 2008. doi: 10.1109/TBME.2008.919128.

[51] Fabio Aloise, Francesca Schettini, Pietro Aricò, F. Leotta, Serenella Salinari, Donatella Mattia, F. Babiloni, and Febo Cincotti. P300-based brain–computer interface for environmental control: an asynchronous approach. *Journal of Neural Engineering*, 8(2):025025, 2011. doi: 10.1088/1741-2560/8/2/025025.

[52] Erik J. Scheme, B. S. Hudgins, and Kevin B. Englehart. Confidence-based rejection for improved pattern recognition myoelectric control. *IEEE Transactions on Biomedical Engineering*, 60(6):1563–1570, 2013. doi: 10.1109/TBME.2013.2238939.

[53] Amy L. Orsborn, Helene G. Moorman, Simon A. Overduin, Maryam M. Shanechi, Dragan F. Dimitrov, and Jose M. Carmena. Closed-loop decoder adaptation shapes neural plasticity for skillful neuroprosthetic control. *Neuron*, 82(6):1380–1393, 2014. doi: 10.1016/j.neuron.2014.04.048.

[54] Beata Jarosiewicz, Anish A. Sarma, Daniel Bacher, Nicolas Y. Masse, John D. Simeral, Brittany Sorice, Erin M. Oakley, Christine Blabe, Chethan Pandarinath, Vikash Gilja, Sydney S. Cash, Emad N. Eskandar, Gerhard Friehs, Jaimie M. Henderson, Krishna V. Shenoy, John P. Donoghue, and Leigh R. Hochberg. Virtual typing by people with tetraplegia using a self-calibrating intracortical brain-computer interface. *Science Translational Medicine*, 7 (313):313ra179, 2015. doi: 10.1126/scitranslmed.aac7328.

[55] Alan D. Degenhart, William E. Bishop, Emily R. Oby, Elizabeth C. Tyler-Kabara, Steven M. Chase, Aaron P. Batista, and Byron M. Yu. Stabilization of a brain–computer interface via the alignment of low-dimensional spaces of neural activity. *Nature Biomedical Engineering*, 4(7):672–685, 2020. doi: 10.1038/s41551-020-0542-9.

[56] Chaofei Fan, Nick Hahn, Foram Kamdar, Donald Avansino, Guy Wilson, Leigh Hochberg, Krishna V. Shenoy, Jaimie Henderson, and Francis Willett. Plug-and-play stability for intracortical brain-computer interfaces: A one-year demonstration of seamless brain-to-text communication. In *Advances in Neural Information Processing Systems*, volume 36, pages 42258–42270, 2023. doi: 10.52202/075280-1831.

[57] Guy H. Wilson, Elias A. Stein, Foram Kamdar, Donald T. Avansino, Tsam Kiu Pun, Ronnie Gross, Tommy Hosman, Tyler Singer-Clark, Anastasia Kapitonava, Leigh R. Hochberg, John D. Simeral, Krishna V. Shenoy, Shaul Druckmann, Jaimie M. Henderson, and Francis R. Willett. Long-term unsupervised recalibration of cursor-based intracortical brain–computer interfaces using a hidden Markov model. *Nature Biomedical Engineering*, 10(7): 1466–1484, 2026. doi: 10.1038/s41551-025-01536-z. Published online 8 December 2025. Preprint: bioRxiv 2023, doi:10.1101/2023.02.03.527022.

[58] Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, D. Sculley, Sebastian Nowozin, Joshua V. Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can you trust your model's uncertainty? evaluating predictive uncertainty under dataset shift. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL https://proceedings.neurips.cc/paper_files/paper/2019/hash/8558cb408c1d76621371888657d2eb1d-Abstract.html. arXiv:1906.02530.

[59] Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *International Conference on Learning Representations (ICLR)*, 2017. arXiv:1610.02136.

[60] Dan Bohus and Alexander I. Rudnicky. A principled approach for rejection threshold optimization in spoken dialog systems. In *Proceedings of Interspeech 2005*, pages 2781–2784, 2005. doi: 10.21437/Interspeech.2005-820.

[61] Anastasios N. Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification, 2021. arXiv:2107.07511.

[62] Ryan J. Tibshirani, Rina Foygel Barber, Emmanuel J. Candès, and Aaditya Ramdas. Conformal prediction under covariate shift. In *Advances in Neural Information Processing*

*Systems*, volume 32, 2019. URL https://proceedings.neurips.cc/paper_files/paper/2019/hash/8fb21ee7a2207526da55a679f0332de2-Abstract.html. arXiv:1904.06019.

[63] Rina Foygel Barber, Emmanuel J. Candès, Aaditya Ramdas, and Ryan J. Tibshirani. Conformal prediction beyond exchangeability. *The Annals of Statistics*, 51(2):816–845, 2023. doi: 10.1214/23-AOS2276. arXiv:2202.13415.

[64] António Farinhas, Chrysoula Zerva, Dennis Ulmer, and André F. T. Martins. Non-exchangeable conformal risk control. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.01262.

[65] Isaac Gibbs and Emmanuel Candès. Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems*, volume 34, pages 1660–1672, 2021. URL https://proceedings.neurips.cc/paper_files/paper/2021/hash/0d441de75945e5acbc865406fc9a2559-Abstract.html. arXiv:2106.00170.

[66] Isaac Gibbs and Emmanuel Candès. Conformal inference for online prediction with arbitrary distribution shifts, 2022. arXiv:2208.08401.

[67] Anastasios N. Angelopoulos, Emmanuel J. Candès, and Ryan J. Tibshirani. Conformal PID control for time series prediction. In *Advances in Neural Information Processing Systems*, volume 36, pages 23047–23074, 2023. doi: 10.52202/075280-1000. arXiv:2307.16895.

[68] Shai Feldman, Liran Ringel, Stephen Bates, and Yaniv Romano. Achieving risk control in online learning settings. *Transactions on Machine Learning Research*, 2023. URL https://openreview.net/forum?id=5Y04GWvoJu. arXiv:2205.09095.

[69] Fares Ernez, Alexandre Arnold, Audrey Galametz, Catherine Kobus, and Nawal Ould-Amer. Applying the conformal prediction paradigm for the uncertainty quantification of an end-to-end automatic speech recognition model (wav2vec 2.0). In *Proceedings of the Twelfth Symposium on Conformal and Probabilistic Prediction with Applications*, volume 204 of *Proceedings of Machine Learning Research*, pages 1–20, 2023.

[70] Victor Quach, Adam Fisch, Tal Schuster, Adam Yala, Jae Ho Sohn, Tommi Jaakkola, and Regina Barzilay. Conformal language modeling. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2306.10193.

[71] Christopher Mohri and Tatsunori Hashimoto. Language models with conformal factuality guarantees, 2024. arXiv:2402.10978.

[72] Yasin Abbasi-Yadkori, Ilja Kuzborskij, David Stutz, András György, Adam Fisch, Arnaud Doucet, Iuliya Beloshapka, Wei-Hung Weng, Yao-Yuan Yang, Csaba Szepesvári, Ali Taylan Cemgil, and Nenad Tomasev. Mitigating LLM hallucinations via conformal abstention, 2024. arXiv:2405.01563.

[73] Chen Chen, Yuchen Hu, Chao-Han Huck Yang, Sabato Marco Siniscalchi, Pin-Yu Chen, and Eng Siong Chng. HyPoradise: An open baseline for generative speech recognition with large language models. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, volume 36, pages 31665–31688, 2023. doi: 10.52202/075280-1375. arXiv:2309.15701.

[74] Wenbin Huang, Yuhang Qiu, Bohan Li, Yiwei Guo, Jing Peng, Hankun Wang, Xie Chen, and Kai Yu. RAS: a reliability oriented metric for automatic speech recognition, 2026. arXiv:2604.24278.

[75] Bernardo Dal Seno, Matteo Matteucci, and Luca T. Mainardi. The utility metric: A novel method to assess the overall performance of discrete brain–computer interfaces. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 18(1):20–28, 2010. doi: 10.1109/TNSRE.2009.2032642.

[76] G. Townsend, B. K. LaPallo, C. B. Boulay, D. J. Krusienski, G. E. Frye, C. K. Hauser, N. E. Schwartz, T. M. Vaughan, J. R. Wolpaw, and E. W. Sellers. A novel P300-based brain–computer interface stimulus presentation paradigm: Moving beyond rows and columns. *Clinical Neurophysiology*, 121(7):1109–1120, 2010. doi: 10.1016/j.clinph.2010.01.030.

[77] R. William Soukoreff and I. Scott MacKenzie. Metrics for text entry research: An evaluation of MSD and KSPC, and a new unified error metric. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI)*, pages 113–120, 2003. doi: 10.1145/642611.642632.

[78] Jonathan R. Wolpaw, Niels Birbaumer, Dennis J. McFarland, Gert Pfurtscheller, and Theresa M. Vaughan. Brain–computer interfaces for communication and control. *Clinical Neurophysiology*, 113:767–791, 2002.

[79] David J. Ward, Alan F. Blackwell, and David J. C. MacKay. Dasher—a data entry interface using continuous gestures and language models. In *Proceedings of the 13th Annual ACM Symposium on User Interface Software and Technology (UIST)*, pages 129–137, 2000. doi: 10.1145/354401.354427.

[80] Heidi Horstmann Koester and Simon P. Levine. Modeling the speed of text entry with a word prediction interface. *IEEE Transactions on Rehabilitation Engineering*, 2(3):177–187, 1994. doi: 10.1109/86.331567.

[81] Oliver Rübel, Andrew Tritt, Ryan Ly, Benjamin K. Dichter, Satrajit Ghosh, Lawrence Niu, Pamela Baker, Ivan Soltesz, Lydia Ng, Karel Svoboda, Loren Frank, and Kristofer E. Bouchard. The Neurodata Without Borders ecosystem for neurophysiological data science. *eLife*, 11:e78362, 2022. doi: 10.7554/eLife.78362.

[82] Gerwin Schalk, Dennis J. McFarland, T. Hinterberger, Niels Birbaumer, and Jonathan R. Wolpaw. BCI2000: A general-purpose brain-computer interface (BCI) system. *IEEE Transactions on Biomedical Engineering*, 51(6):1034–1043, 2004. doi: 10.1109/TBME.2004.827072.

[83] Yann Renard, Fabien Lotte, Guillaume Gibert, Marco Congedo, Emmanuel Maby, Vincent Delannoy, Olivier Bertrand, and Anatole Lécuyer. OpenViBE: An open-source software platform to design, test, and use brain–computer interfaces in real and virtual environments. *Presence: Teleoperators and Virtual Environments*, 19(1):35–53, 2010. doi: 10.1162/pres.19.1.35.

[84] IEEE. P2731: IEEE draft standard for a unified terminology for brain-computer interfaces. IEEE Standards Association project page, 2018. URL https://standards.ieee.org/ieee/2731/7383/. Active PAR, approved 27 September 2018.

[85] IEEE. P2794: IEEE draft trial-use standard for reporting standards for in vivo neural interface research (RSNIR). IEEE Standards Association project page, 2022. URL https://standards.ieee.org/ieee/2794/11108/. Active PAR, approved 3 December 2022.

[86] Calvin Eiber, Jean Delbeke, Jorge Cardoso, Martijn de Neeling, Sam John, Chang Won Lee, Jerry Skefos, Argus Sun, Dimiter Prodanov, and Zach McKinney. Preliminary minimum reporting requirements for in-vivo neural interface research: I. implantable neural interfaces.

*IEEE Open Journal of Engineering in Medicine and Biology*, 2:74–83, 2021. doi: 10.1109/OJEMB.2021.3060919.

[87] Vinay Jayaram and Alexandre Barachant. MOABB: Trustworthy algorithm benchmarking for BCIs, 2018. arXiv:1805.06427.

[88] Hubert Banville, Stéphane d'Ascoli, Simon Dahan, Jérémy Rapin, et al. NeuralBench: A unifying framework to benchmark NeuroAI models, 2026. arXiv:2605.08495.

[89] José Zariffa. Electromyographic typing gesture classification dataset for neurotechnological human-machine interfaces. Borealis (University of Toronto Dataverse), version 1.0, 2024. CC BY 4.0.

[90] BCBL. DECOMEG — brain activity during typing (MEG & EEG) (dataset). Hugging Face, https://huggingface.co/datasets/bcbl190626/SpanishBCBL, 2026. Basque Center on Cognition, Brain and Language. CC BY-NC 4.0.

[91] Anmol Gulati, James Qin, Chung-Cheng Chiu, Niki Parmar, Yu Zhang, Jiahui Yu, Wei Han, Shibo Wang, Zhengdong Zhang, Yonghui Wu, and Ruoming Pang. Conformer: Convolution-augmented transformer for speech recognition, 2020. arXiv:2005.08100.

# A Notation

**Table 4: Notation.** Prose uses "error among commits"; tables use EAC.

| Symbol | Meaning | Symbol | Meaning |
|---|---|---|---|
| $F_i$, $V$ | frames in the stream of item $i$; vocabulary size | $\lambda$ | commit threshold on $s_i$ |
| $K$ | candidate list size ($K = 10$; 11 for the published gate) | $s_i$ | commit score: $\hat{p}_{\text{top1}}$ if eligible, else $-\infty$ |
| $\hat{p}_{\text{top1}}$ | calibrated probability of the top candidate | $c_i$ | 1 if the top candidate is wrong |
| $\tau_f$, $\tau_s$ | frame and sentence temperatures | $L_i(\lambda)$ | loss $c_i\,\mathbf{1}[s_i \geq \lambda]$ |
| $g_i$ | guard risk of the top candidate, in $[0, 1]$ | $R(\lambda)$, $\hat{R}_n(\lambda)$ | population and calibration risk |
| $r_{\text{abst}}$, $r_{\max}$ | guard limits for abstain and commit | $n$ | number of calibration items |
| $k$, $k_{\max}$ | offered candidates; offer limit | $\alpha$ | target wrong-commit rate (0.05) |
| $q$ | conformal quantile of the sentence set | $\beta$ | target error among commits (0.05, 0.10, 0.20) |
| $T$, $d$ | items in an online sequence; feedback delay | $\delta$ | failure probability of LTT and SGR (0.1) |
| $\eta$, $\gamma$ | step sizes of the online controllers | $N$, $\rho$ | window size and decay of online CRC |

# B Algorithms

Algorithm 1 is the decision for one item, and Algorithm 2 the fitting and calibration. Grid values are in Table 8; the reference implementation is `fidetype.trust` and `fidetype.eval.harness`.

**Algorithm 1** Trust decision for one item

**Require:** stream $\log p_t(v)$, $t = 1..F$; fitted $\tau_f, \tau_s, q$; policy $(\lambda, r_{\max}, r_{\text{abst}}, k_{\max}, \text{fallback})$

$\tilde{\ell}_t(v) \leftarrow \log p_t(v)/\tau_f - \log \sum_u p_t(u)^{1/\tau_f}$ ▷ frame calibration, before the LM

$(y_1..y_K, \text{score}_1..\text{score}_K) \leftarrow$ CTC prefix beam search with LM on $\tilde{\ell}$ ▷ $K$-best list

$\hat{p}_j \leftarrow \text{softmax}_j(\text{score}/\tau_s)$; $\quad \hat{p}_{\text{top1}} \leftarrow \hat{p}_1$

$C \leftarrow \{\, j : 1 - \hat{p}_j \le q \,\}$, ordered by $\hat{p}_j$ ▷ conformal sentence set

**for** each character $m$ of $y_1$ **do** ▷ acoustic confidence

  char streams: $a_m \leftarrow \max_{t \in \text{span}(m)} \exp \tilde{\ell}_t(y_{1,m})$ (CTC forced alignment of $y_1$)

  phoneme path: $a_m \leftarrow 0.5^e \min_{\text{phonemes of the word}} \exp \tilde{\ell}$ at their peak frames ($e$ = lexicon edits)

**end for**

$\text{low} \leftarrow \frac{1}{|y_1|} \#\{m : a_m < 0.5\}$ $\quad$ (low $\leftarrow 1$ if $y_1$ is empty)

$\text{cons} \leftarrow \sum_{j \le 5} \hat{p}_j \, \text{sim}(y_1, y_j) / \sum_{j \le 5} \hat{p}_j, \quad \text{sim}(a, b) = 1 - \text{WED}(a, b)/\max(|a|_w, |b|_w)$

$g \leftarrow \min\big(1, \max(\text{low}, 1 - \text{cons})\big)$ ▷ guard risk; WED = word edit distance

**if** $g \ge r_{\text{abst}}$ **then return** ABSTAIN

**end if**

$s \leftarrow \hat{p}_{\text{top1}}$ if $g \le r_{\max}$ else $-\infty$ ▷ commit score

**if** $s \ge \lambda$ **then return** COMMIT $y_1$

**end if**

**if** $1 \le |C| \le k_{\max}$ **then return** OFFER $C$

**end if**

**return** OFFER $y_1..y_{k_{\max}}$ if fallback = offer and $k_{\max} \ge 1$, else ABSTAIN

**Algorithm 2** Fitting and calibration of the commit threshold

**Require:** fit items $\mathcal{F}$ (split into $\mathcal{F}_1$, $\mathcal{F}_2$, $\mathcal{F}_3$; see below), disjoint calibration items $\mathcal{C}$ ($n$ items with truth), $\alpha$, $\delta$

$\tau_f \leftarrow \arg\min_\tau$ cross-entropy of the aligned true tokens at the greedy CTC peaks of $\mathcal{F}_1$ ▷ golden section, $\tau \in [0.05, 20]$

$\tau_s \leftarrow \arg\min_\tau$ binary NLL of $[y_1 = \text{truth}]$ under $\text{softmax}(\text{score}/\tau)_1$ on $\mathcal{F}_1$

$q \leftarrow$ the $\lceil (m+1)(1-0.1) \rceil$-th smallest $1 - \hat{p}_{\text{truth}}$ over the $m$ items of $\mathcal{F}_2$ whose truth is in the $K$-best list

$(\lambda_{\text{tuned}}, r_{\max}, r_{\text{abst}}, k_{\max}, \text{fallback}) \leftarrow$ grid point with the highest eWPM on $\mathcal{F}_3$ among those with wrong-commit rate $\le \alpha$ (first in grid order on ties; if none, the lowest wrong-commit rate)

compute $s_i$ and $c_i$ for every $i \in \mathcal{C}$ (Alg. 1 up to the commit score)

**CRC:** $\hat{\lambda} \leftarrow \min\{\lambda \in \{s_i\} \cup \{+\infty\} : (\sum_i c_i \mathbf{1}[s_i \ge \lambda] + 1)/(n+1) \le \alpha\}$

**LTT:** for $\lambda = 1.00, 0.99, \ldots, 0.00$: $k \leftarrow \sum_i c_i \mathbf{1}[s_i \ge \lambda]$; if $\Pr(\text{Bin}(n, \alpha) \le k) \le \delta$ then $\hat{\lambda} \leftarrow \lambda$ else stop ▷ $\hat{\lambda} = +\infty$ if the first test fails

**SGR** (target $\beta$): bisection over the sorted finite $s_i$ ($m$ eligible items, $\lceil \log_2 m \rceil$ steps); accept $\lambda$ if the exact binomial upper $(1 - \delta/\lceil \log_2 m \rceil)$ bound on $\Pr(c = 1 \mid s \ge \lambda)$ is $\le \beta$; $\hat{\lambda}_{\text{SGR}} \leftarrow$ the lowest accepted $\lambda$ ($+\infty$ if none)

**return** $\lambda \leftarrow \max(\lambda_{\text{tuned}}, \hat{\lambda})$ ▷ max rule; joint: $\max(\lambda_{\text{tuned}}, \hat{\lambda}_{\text{CRC}}, \hat{\lambda}_{\text{SGR}})$

Fit sets. Brain-to-Text '25: $\mathcal{F}_1$ = first half of the fit items in file order, $\mathcal{F}_2$ = second half, $\mathcal{F}_3$ = all fit items. sEMG: $\mathcal{F}_1$, $\mathcal{F}_2$ = the two halves of fit half A; $\mathcal{F}_3$ = fit half B (or its policy part). MEG: $\mathcal{F}_1$, $\mathcal{F}_2$ = halves of the validation instances, $\mathcal{F}_3$ = all validation instances. Each item has one attempt, so the policy always uses its final-attempt fallback. LTT-sel replaces the LTT test by $\Pr(\text{Bin}(n_\lambda, \beta) \le k_\lambda) \le \delta/20$ over the grid $0.05, 0.10, \ldots, 1.00$, where $n_\lambda$ is the number of calibration commits.

# C Decision flow

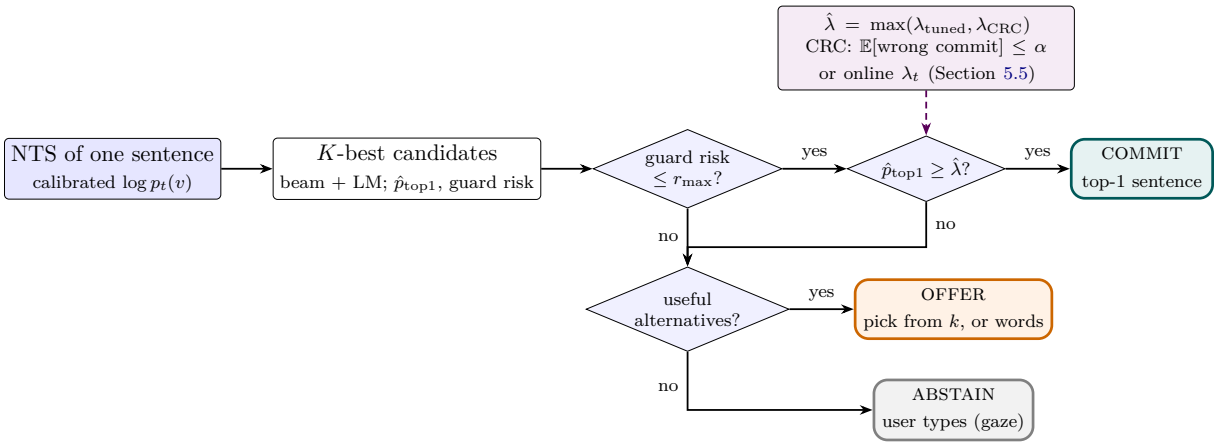



**Figure 4: Trust decision for one item.** Candidates come from the calibrated stream. A candidate is committed only if the neural-evidence guard accepts it and its calibrated confidence reaches the threshold $\hat{\lambda}$, which is the larger of the eWPM-tuned threshold and the CRC threshold (or the online threshold $\lambda_t$). Otherwise the layer offers alternatives or abstains, and the user types with the fallback keyboard.

# D User model, variants and baselines

Table 5 lists the parameters of the main simulated gaze-typing user (word model: the user fixes single words). Only the fallback typing speed and the per-slot scan-and-select time are matched to measurements of one eye-gaze user with ALS [28]; all other values are assumptions.

**Variants.** Each variant changes one group of parameters and refits every policy: fallback speed 10 or 40 characters per minute (main 20); all selection, scanning, word-selection and navigation times scaled by 0.5 or 2; miss probability of a wrong committed word 0 or 0.2 (main 0.05). In the *symmetric-offer* variants, an offer without the intended sentence makes the user fix the closest candidate word by word, missing each wrong word with the same probability as after a commit; there is no oracle choice between fixing, repeating and retyping, and full retyping is disabled for all actions. In the *misspecified* evaluation, the trust and always-offer policies fitted under the main user are scored under each variant without refitting.

**Gaze typing with word completion.** The user types letter by letter at the fallback speed. Before each letter, the keyboard shows the three most frequent vocabulary words (counts $\geq 2$) that start with the typed prefix; there is no sentence context. If the intended word is shown and selecting it is faster than typing the rest of the word and its space, the user selects it at the scan-and-select cost for three slots plus the word-selection time (4.5 s at the main costs, close to the 4.7 s measured for one word choice [28]). The main baseline charges no time for looking at the list, which favours it; the *glance* variant charges 1 s per look. Unigram corpora: Brain-to-Text '25, the word unigram of the decoder (Tatoeba English); sEMG, Tatoeba English without the held-out test pool; MEG, Tatoeba Spanish without the SpanishBCBL sentences.

**Table 5: Main simulated-user parameters.**

| Parameter | Value | Basis |
|---|---|---|
| Fallback (gaze keyboard) speed | 20 characters/min | measured, eye-gaze user LP1 [28] |
| Scan and select one of $k$ word alternatives | $1.0 + 1.0 \log_2(k+1) + 1.5$ s (4.5 s for $k = 3$) | sum matched to 4.7 s [28] |
| Select one of $k$ offered sentences | $1.0 + 0.5 \log_2(k+1)$ s | assumption |
| Reading time | 0.3 s per word, 0.5 s per candidate | assumption |
| Navigate to and delete a wrong word | 3.0 s (2 selections) | assumption |
| Miss probability of a wrong, unmarked word (commit, OFFER_WORDS) | 0.05 | assumption (no measurement) |
| Errors in offered candidates | always seen (main model) | assumption |
| Full retype overhead | 1.5 s; chosen when cheaper than fixing words | assumption (rational user) |
| Abstain overhead | 0.5 s, then fallback typing | assumption |
| Neural input speed, MEG / B2T '25 / sEMG | 152 / 159.3 / 75.8 characters/min | device setting |

# E Decoder quality

Table 6 gives the decoder metrics and input controls. Excluding the 6 MEG test sentences with a close training variant (CER $\leq 0.25$ to a training sentence) gave 0.519 (greedy) and 0.492 (beam), so memorisation of near variants explains at most 0.02 of the CER. On sEMG, plain key accuracies (0.299, 0.168, 0.209) are below the majority-class rate (space, 0.338) because the CNNs rarely predict space; balanced accuracy is the fair metric.

**Table 6: Decoder quality and input controls (test sets).** Exact: share of items whose top-1 sentence is right; in $K$-best: share whose intended sentence is in the 10-best list. MEG: the trust harness keeps an insertion bonus of 1.0 tuned on raw streams; the validation-tuned bonus of 0 gives the 0.475 row (Section 6). sEMG chance: majority class 0.338, uniform 0.037. "—": not applicable or not computed.

| Device / decoder | CER | WER | Exact / in $K$-best | Controls and floors |
|---|---|---|---|---|
| MEG, asynchronous, greedy | 0.506 | 0.842 | — | shuffled 0.760, zeros 0.925 |
| MEG, beam + LM tuned on validation | 0.475 | 0.809 | 0.005 / — | text-only floor 0.688 |
| MEG, beam in the trust harness | 0.522 | — | 0.003 / 0.016 | — |
| B2T '25, our phoneme-to-word decoder | 0.095 | 0.149 | 0.390 / 0.623 | — |
| B2T '25, published decoder | 0.031 | 0.043 | 0.806 / — | — |
| sEMG keys: accuracy (balanced) | 0.299 (0.367) / 0.168 (0.240) / 0.209 (0.226) | | — | permuted labels 0.022 (0.032) |
| sEMG sentences after LM | 0.123 / 0.398 / 0.479 | 0.214 / 0.579 / 0.681 | 0.459 / 0.118 / 0.038 | in $K$-best 0.64 / 0.21 / 0.08 |

**Table 7: MEG decoder over three training seeds** (637 test instances; trust harness with the earlier settings, fitted on validation). Seed 0 is the paper's decoder; only the seed differs. Trust − gaze: two-way bootstrap of the trust harness (seed 0 in the unified bootstrap: [−0.10, +0.25]).

| | Seed 0 | Seed 1 | Seed 2 | Mean ± SD |
|---|---|---|---|---|
| Test CER, greedy | 0.506 | 0.502 | 0.592 | $0.533 \pm 0.051$ |
| Test CER, beam + LM (validation-tuned) | 0.475 | 0.467 | 0.567 | $0.503 \pm 0.056$ |
| Shuffled MEG, greedy | 0.760 | 0.745 | 0.732 | $0.746 \pm 0.014$ |
| Zero input, greedy | 0.925 | 0.948 | 0.819 | $0.897 \pm 0.069$ |
| Always commit: eWPM / wrong-commit | 2.58 / 0.997 | 2.55 / 0.994 | 2.35 / 1.000 | |
| Trust: eWPM / commits | 3.20 / 0 | 3.16 / 0 | 2.96 / 0 | |
| Trust − gaze (gaze 3.15) | [−0.11, +0.26] | [−0.15, +0.23] | [−0.30, −0.05] | |
| Eligible to commit / truth in 10-best | 0.024 / 0.016 | 0.036 / 0.019 | 0.002 / 0.000 | |

# F Per-device settings

Table 8 lists the parameters that the shared implementation fitted or used per device. The policy grid has 1,980 points: commit threshold $\lambda \in \{0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.85, 0.9, 0.95, 0.98, 1.01\}$, maximum guard risk for a commit $r_{\max} \in \{0.1, 0.2, 0.3, 0.5, 1.0\}$, offer limit $k_{\max} \in \{0, 1, 2, 3, 5, 8\}$, abstain risk $r_{\text{abst}} \in \{0.6, 0.8, 1.01\}$ and fallback $\in$ {offer, abstain}. Decoder settings are in the experiment configurations in the repository.

**Table 8: Per-device settings of the shared trust-layer implementation.** Fitted on each device's fit data (MEG: validation set). $\tau_s = 20.0$ is the upper end of the search interval. Max. guard risk: largest guard risk that still allows a commit.

| | MEG | B2T '25, ours | B2T '25, published gate | sEMG (cal. / x-sess. / x-subj.) |
|---|---|---|---|---|
| Frame temperature $\tau_f$ | 3.30 | 3.78 | 3.78 | 1.25 / 1.78 / 2.17 |
| Sentence temperature $\tau_s$ | 20.0 | 4.56 | 1.72 | 2.42 / 5.42 / 20.0 |
| LM weight (beam) | 0.5 (insertion bonus 1.0) | 1.0 (word LM; edit penalty 12, word bonus 3) | as ours | 1.0 / 1.0 / 1.5 |
| Candidate list $K$ | 10 | 10 | published sentence + our 10 (bonus +2.0) | 10 |
| Plug-in commit threshold | 0.5 | 0.5 | 0.7 | 0.7 / 0.3 / 0.3 |
| Max. guard risk for commit | 0.3 | 0.5 | 0.5 | 1.0 / 1.0 / 0.1 |
| Offer size $k$ | 1 | 2 | 1 | 2 / 2 / 1 |
| Fallback after offer | offer | offer | offer | offer |
| Guard | low-evidence threshold 0.5; consistency over the top 5; risk = max(low evidence, 1 − consistency), weights 1 | | | |

# G All policies on the fixed test sets

Table 9 extends Table 1 with the plain threshold, pure CRC and pure LTT (which fill the risk budget), and the LM weight re-tuned on sEMG cross-subject. On sEMG and MEG, pure CRC commits items that the plug-in rule rejects, which raises the wrong-commit rate ten-fold on sEMG and commits mostly wrong text on MEG; the max rule prevents this.

**Table 9: All policies on the fixed test sets.** Columns as in Table 1. Plain threshold: calibrated $\hat{p}_{\text{top1}}$, no guard (= trust on sEMG and MEG). Pure CRC / LTT: fit-A pipeline, clean calibration (B2T split A, $n = 401$; sEMG 143; MEG 199 instances of 5 sentences, no guarantee for new sentences), without the max rule. Unified bootstrap ($B = 2000$, seed 20261112).

| Device | Policy | eWPM [CI] | Wrong-commit [CI] | Commit | EAC | Silent |
|---|---|---|---|---|---|---|
| B2T '25, ours | plain threshold | 11.51 [10.79, 12.35] | 0.036 [0.025, 0.046] | 0.17 | 0.22 | 0 |
| | CRC, pure (split A) | 11.48 [10.79, 12.29] | 0.025 [0.015, 0.035] | 0.10 | 0.24 | 0.001 |
| | LTT, pure (split A) | 11.48 [10.79, 12.29] | 0.027 [0.016, 0.038] | 0.10 | 0.26 | 0.001 |
| B2T '25, published | plain threshold | 18.62 [17.01, 20.17] | 0.025 [0.014, 0.038] | 0.38 | 0.06 | 0.001 |
| | CRC, pure (split A) | 18.79 [17.13, 20.37] | 0.048 [0.029, 0.071] | 0.49 | 0.10 | 0.005 |
| | LTT, pure (split A) | 18.65 [17.04, 20.20] | 0.029 [0.013, 0.050] | 0.40 | 0.07 | 0.001 |
| sEMG session-cal. | CRC, pure | 8.07 [6.98, 9.26] | 0.040 [0.025, 0.056] | 0.24 | 0.17 | 0.005 |
| | LTT, pure | 8.04 [6.97, 9.21] | 0.022 [0.009, 0.036] | 0.18 | 0.12 | 0.002 |
| sEMG cross-session | always offer | 4.67 [4.19, 5.25] | 0 | 0 | — | 0 |
| | CRC, pure | 4.66 [4.18, 5.24] | 0.056 [0.039, 0.074] | 0.09 | 0.61 | 0.007 |
| | LTT, pure | 4.67 [4.18, 5.25] | 0.031 [0.021, 0.042] | 0.05 | 0.57 | 0.005 |
| sEMG cross-subject | LM weight re-tuned | 4.84 [4.28, 5.51] | 0.002 [0, 0.006] | 0.01 | 1/6 | 0 |
| MEG | CRC, pure (instance level) | 3.20 [3.01, 3.43] | 0.028 [0.006, 0.061] | 0.03 | 18/19 | 0.005 |
| | LTT, pure (instance level) | 3.20 [3.01, 3.44] | 0 | 0.002 | 0/1 | 0 |

# H Ablation

Table 10 refits each variant on the condition's fit data and scores it on the same test items. *No calibration*: no temperature. *Confidence only*: both temperatures fitted, but the beam decodes the raw stream. *No sentence temperature*: the frame temperature is applied before the beam and the sentence calibrator is the identity. *No conformal sets*: OFFER shows the fitted top $k$. The OFFER_WORDS rows use a smaller common grid so that they are comparable. *LM weight re-tuned*: on sEMG, the LM weight is chosen on fit-A CER with the frame-calibrated streams (cross-session 1.0 to 0.5, cross-subject 1.5 to 0.5); on MEG, the validation-tuned beam (LM weight 0.25, insertion bonus 0). Top-1 CER on sEMG cross-subject: 0.479 (trust), 0.369 (raw decoding), 0.360 (LM weight re-tuned).

**Table 10: Unified ablation** (eWPM / wrong-commit rate; point values; "=": equal to trust). CIs and paired differences at the primary unit (unified bootstrap) are in the repository (`unified-ci/results/rerun/ablation`).

| Variant | B2T '25 ours | B2T '25 published | sEMG s.-cal. | sEMG x-sess. | sEMG x-subj. | MEG |
|---|---|---|---|---|---|---|
| trust (as reported) | 11.55 / 0.059 | 18.77 / 0.044 | 8.02 / 0.004 | 4.68 / 0.004 | 4.08 / 0 | 3.20 / 0 |
| no calibration | 11.57 / 0.071 | 18.73 / 0.043 | 8.15 / 0.045 | 4.75 / 0.013 | 4.76 / 0.005 | 3.04 / 0.005 |
| confidence only (raw decoding) | 11.54 / 0.074 | 18.74 / 0.047 | 8.07 / 0.018 | 4.76 / 0.007 | 4.75 / 0 | 3.04 / 0 |
| no sentence temperature | 11.50 / 0.076 | 18.66 / 0.032 | 8.11 / 0.040 | 4.67 / 0.018 | 4.07 / 0.044 | 3.20 / 0 |
| no guard | 11.51 / 0.036 | 18.62 / 0.025 | = | = | = | = |
| no conformal sets | 11.53 / 0.059 | = | 8.07 / 0.020 | = | = | = |
| OFFER_WORDS off (small grid) | 11.25 / 0.032 | 18.62 / 0.025 | 7.98 / 0.004 | 4.64 / 0 | 4.08 / 0 | 3.20 / 0 |
| OFFER_WORDS on (small grid) | 11.32 / 0.054 | 18.59 / 0.026 | 7.99 / 0.015 | 4.64 / 0.005 | 4.08 / 0 | 3.20 / 0 |
| LM weight re-tuned | — | — | 8.02 / 0.004 | 4.73 / 0.013 | 4.84 / 0.002 | 3.14 / 0 |
| CRC, clean cal. (wrong-commit, off → on) | 0.025 → 0.025 | 0.048 → 0.048 | 0.004 → 0.040 | 0.004 → 0.056 | 0 → 0 | 0 → 0.028 |
| gaze typing only | 4.08 | 4.08 | 4.27 | 4.27 | 4.27 | 3.15 |

# I LM leakage check

A near variant is a sentence within a character edit distance of 0.25 times its length, after normalisation, in the 600,000-sentence training slice of the English character LM. Of the 1,717 unique Brain-to-Text '25 cue sentences, 22 have an exact copy and 310 an exact or near variant; of the 551 sEMG test sentences, none has an exact copy and 295 have a near variant (Tatoeba is heavily templated). We rebuilt the LMs without these sentences (849 removed for Brain-to-Text '25, plus 910 from the word unigram; 2,895 for sEMG, including near variants of the fit sentences) and re-ran the trust harness. On Brain-to-Text '25, 2 of 1,628 top-1 sentences changed and all main numbers were identical to three decimals. On sEMG, trust eWPM changed from 8.02, 4.68 and 4.08 to 7.98, 4.65 and 4.09 (paired participant CIs $[-0.08, -0.01]$, $[-0.05, -0.01]$, $[-0.01, +0.03]$), and no wrong-commit rate changed by more than 0.002. The re-tuned LM weights were unchanged.

**Test-set exclusion (a different analysis).** Rebuilding the LM does not remove the advantage of templated test sentences, which are easier to decode. We therefore also scored every policy on the 256 sEMG test sentences without a near variant, with the original LM. Trust eWPM fell from 8.02, 4.68 and 4.08 to 6.97 [5.94, 8.09], 4.08 [3.77, 4.42] and 3.69 [3.50, 3.92]; on the 295 near-variant sentences alone it was 9.31, 5.40 and 4.52. Trust minus gaze typing became $[+1.82, +3.92]$, $[-0.37, +0.27]$ and $[-0.66, -0.23]$, so on the cross-subject decoder gaze typing is faster than the layer on non-templated text. Always-commit wrong-commit rates rose from 0.541, 0.882 and 0.962 to 0.664, 0.961 and 0.992. The layer stayed safe: 1 wrong commit in 256 sentences (session-calibrated; one-sided upper bound 0.018) and none in the other conditions; pure CRC gave 0.047 and 0.051, and the max rule equals the plug-in rule. On Brain-to-Text '25, excluding the 147 test trials with a near variant (669 kept) gave a trust wrong-commit rate of 0.055 (all: 0.059) on our decoder and 0.042 (0.044) on the published gate. We report all 551 sEMG sentences as the primary population because the held-out pool was fixed before evaluation, and the 256-sentence rows as a sensitivity analysis (Table 1).

# J Datasets and licences

**Table 11: Datasets, resources and licences.**

| Dataset / resource | Licence | Use | Note |
|---|---|---|---|
| SpanishBCBL [6, 90] | CC BY-NC 4.0 | MEG decoder training and evaluation | Research only; weights trained on it are not released for commercial use |
| Brain-to-Text '25 copy-task logits [2, 34] | CC0 1.0 | intracortical trust evaluation | Companion code has no licence; we wrote our own loader |
| Toronto sEMG typing [9, 89] | CC BY 4.0 | sEMG decoder and trust evaluation | Baseline code (CC BY-SA 4.0) not used |
| Tatoeba English / Spanish sentences | CC BY 2.0 FR (some CC0) | character LMs; composed sEMG sentences | Sentences from Tatoeba (`https://tatoeba.org`), by the Tatoeba contributors |
| CMUdict (commit 7479086) | BSD-2-Clause style | phoneme-to-word lexicon | Copyright notice kept |
| Brain2Qwerty code (Meta) | CC BY-NC 4.0 | read only, not copied | Our decoders are our own implementations |
| Fidetype code and specification | Apache-2.0 | all methods | — |

# K MEG decoder details

The asynchronous MEG decoder is a convolutional front end (320 channels) followed by 4 Conformer layers of width 256 [91], trained with CTC [35] and a keystroke auxiliary loss that uses key times as training labels (key pretraining, then CTC training on all 80 recording blocks). At inference, the forward pass takes the MEG segment, its length and the participant index; it has no key input. Table 12 compares the inference inputs with the published asynchronous decoder. We selected the training schedule by validation CER.

**Table 12: Asynchronous setting: our decoder vs Brain2Qwerty v2.**

| Input at inference | Ours | v2 async [7] |
|---|---|---|
| MEG of the sentence segment | yes | yes |
| Segment start | first key − 0.4 s | first key − 400 ms |
| Segment end | last key press + 0.5 s | last key release + 400–500 ms |
| Per-key times or key count | no | no |
| Participant index | yes | yes |
| Key times as training labels | yes (auxiliary loss) | no |

**Leakage checks.** The split is by a hash of the sentence text, and no test sentence occurs in the LM training text. A test sentence is "close" if a training sentence is within CER 0.25; 6 of the 17 test sentences are close. Without them, the CER was 0.519 (greedy) and 0.492 (beam).

**Sentence-grouped calibration.** Instance-level CRC on MEG assumes that calibration and new instances are exchangeable, but instances of one sentence are strongly dependent. A guarantee for new sentences needs the sentence as the unit: group-level CRC needs at least $1/\alpha - 1 = 19$ calibration sentences and LTT 45, while the data set has 5 clean calibration, 11 validation and 17 test sentences (and 19 participants), so group-level CRC was infeasible in every split and

never committed. In 1,000 random splits with all instances of 5, 8 or 11 test sentences as calibration and the other sentences as test, instance-level CRC gave mean test wrong-commit rates of 0.016, 0.018 and 0.019 (95th percentiles 0.025, 0.031 and 0.038; never above $\alpha$), close to random instance splits of the same size (0.017, 0.019, 0.019); participant-grouped splits (6 and 12 of 19 participants) gave 0.015 and 0.019. Only 2.4% of instances were eligible, and 0.91–0.94 of the commits were wrong; the max rule committed nothing. Over seeds, sentence-grouped CRC with 11 calibration sentences gave 0.019, 0.032 and 0.000 (seeds 0, 1, 2).

**Synchronous control decoder.** A synchronous decoder with oracle key times (greedy CER 0.508, beam + LM 0.447; harness top-1 CER 0.451) gave the same trust results as the asynchronous decoder: always commit 2.714, trust 3.229 and gaze typing 3.155 eWPM, trust minus gaze [−0.02, +0.18] over sentences and [−0.07, +0.26] two-way. Over trials, this difference excluded 0 ([+0.03, +0.12]): trial-level intervals make the MEG result look like a gain, and sentence-level intervals do not.

# L Format details and audit log

**Table 13: Devices, data and streams.** All datasets are public. Test units are those of the trust evaluation.

| | **Intracortical (speech)** | **Forearm sEMG (keys)** | **MEG (typing)** |
|---|---|---|---|
| Dataset (licence) | B2T '25 copy task (CC0) | Toronto sEMG (CC BY 4.0) | SpanishBCBL (CC BY-NC 4.0) |
| Participants | 1 (T15, ALS) | 19 (healthy typists) | 19 (healthy typists) |
| Test unit | 816 trials, 14 days | 551 sentences composed from isolated presses | 637 instances of 17 unique sentences |
| Decoder | published online RNN; own phoneme-to-word decoder | own CNN | own asynchronous ConvCTC |
| Stream vocabulary | `phoneme` (CTC, 41 tokens) | `key` (W3C codes) | `char` (CTC) |
| Frame | 80 ms | one per key press | 40 ms |
| Commit decision tested? | yes | weakly (0–9% committed) | no (no right top-1) |

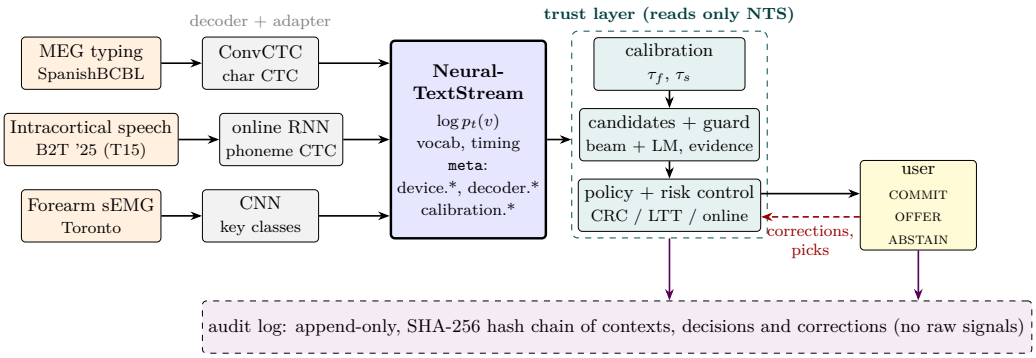



**Figure 5: Architecture.** Each device's decoder emits a stream through a small adapter. The trust layer reads only the stream: it calibrates confidence, builds candidate sentences, and chooses an action under a risk-control rule (Algorithms 1–2). The user's corrections and picks (dashed) can feed online recalibration. Decisions can be written to a hash-chained audit log that stores no raw signals (Appendix L).

**Serialisations and conformance.** The reference serialisations of NeuralTextStream are an NPZ file and a JSON Lines streaming form. A conformance suite runs 60 checks, each tied to a specification clause, and reports one of three cumulative levels: *core* (a consumer must accept the stream), *trust-ready* (modality, causality, calibration state and language are declared) and *provenance* (stream id, decoder name, version and weights hash). It ships 52 golden test vectors (32 NPZ, 20 JSON Lines); the reference consumer passes 52 of 52. On synthetic inputs,

our MEG and sEMG adapters reach the provenance level and the intracortical adapter reaches trust-ready, because the published online decoder has no weights file to hash. Writing the suite found five bugs in our own producers and ten ambiguities in the specification text, now listed as open questions.

**Fields and rules.** Besides the log-probability array, frame times, vocabulary, vocabulary type and blank index, optional fields give the frame period, latency, producer and a `meta` object with named key prefixes (`device.*`, `decoder.*`, `lm.*`, `calibration.*`, `adapter.*`). A field is interpreted as a probability only if the stream declares it as one in `calibration.*`. Raw signals and direct identifiers stay out of the stream (privacy rules).

**Adapters.** The adapters have 412 (intracortical), 568 (sEMG) and 156 (MEG) physical lines of Python, including blank lines and docstrings. Mapping the intracortical and sEMG decoders into the stream needed no change to the frame core of v0.1, a Development release written by the same authors; the trust harness added one optional field (the truth in the stream's token space) and provenance keys.

**Audit log.** Each decision can be appended to a JSON Lines log. Context entries are content-addressed (their id is the SHA-256 of the context) and hold the adapter, decoder (name, version, weights hash, training data and licence), LM, calibration (temperatures, risk-control method, $\alpha$, $\delta$, threshold, $n$) and policy versions; decision entries hold the action, scores, a hash of the stream and its frame range; correction entries link to the decision they correct. Each entry stores the hash of the previous entry and its own SHA-256 over the canonical JSON, so any edit, deletion or reordering breaks the chain; a separately stored head anchor detects truncation. The log stores no raw signals (numeric arrays longer than 64 values are rejected), stores text as a keyed hash by default, and pseudonymises subject and session ids. It is tamper-evident, not tamper-proof: there is no signature in v0.

**Replay.** We replayed the Brain-to-Text '25 test half (816 trials) with the log attached, for the plug-in gate and a CRC gate. The CRC gate of this replay is the earlier one, calibrated on the fit data that also fitted the pipeline (reused data; 160 commits, wrong-commit rate 0.048), not the clean CRC of Table 16; the replay checks the log, not the rule. The log held 1,723 entries and passed verification. It recorded the same wrong commits as the harness: 48 of 48 (plug-in) and 39 of 39 (CRC), of which 2 and 1 were silent. This is a consistency check of the replay: the corrections come from the known cue sentences, and in live use silent errors would be missing from the log. Every tampering case we tried was detected: editing an entry, editing with re-hashing, deleting, reordering, and cutting the tail (with the head anchor). The cost was about 0.1 ms and about 1.1 kB per decision. Figure 6 shows one entry of the audit report.

**Wrong commit** `dec-000041-ed60b4bd` (entry 44, hash `8cf2c6c67053b185...`)
Committed: "who you never expect to when"
Correction (known cue sentence, hash-linked): "who you never expect to win"; 2 character edits, 1 word edit
Scores: $\hat{p}_{\text{top1}}$ 0.5814, set size 3, guard risk 0.2857; reason: $\hat{p}_{\text{top1}} \geq 0.50$
Input: stream `1b4b60e8e4950710...`, frames [0, 335], item `b2t25-test-trial-196`
Decoder: published online RNN + our phoneme-to-word decoder; LM: Tatoeba character 6-gram + word unigram + CMUdict; adapter `fidetype.adapters.b2t25`
Calibration `cal-692fb52a73417552`: $\tau_f$ 3.7809, $\tau_s$ 4.5551, fit set: first half of each day (812 trials)
Policy `pol-2eaa4ff51919527a`: commit threshold 0.5, maximum guard risk 0.5, offer size 2

**Figure 6: One entry of the audit report** (B2T '25 replay, plug-in gate; public CC0 copy-task cue, not the participant's own words; the CC0 dedication allows item-level quotation). Each wrong commit is traced to the exact stream segment and to the decoder, LM, calibration and policy versions that produced it. Hashes are shortened.

# M Online re-calibration methods

**Table 14: Online re-calibration methods for the commit threshold.** $L_t$: realised loss of item $t$; $T$: number of items in the sequence; $d$: feedback delay in items. The two `aci` variants bound the long-run average of the *observed* losses for any sequence of items, without exchangeability; `window` and `decay` have no guarantee under shift.

| Method | Update | Guarantee | Feedback |
|---|---|---|---|
| `window` | CRC on the last $N = 200$ items with known truth | none under shift[a] | full |
| `decay` | weighted CRC on all past items, weights $\rho^{\text{age}}$, $\rho = 0.99$ | none under shift[a] | full |
| `aci` | $\lambda_{t+1} = \max(0, \lambda_t + \eta(L_t - \alpha))$, $\eta = 0.05$ | $\frac{1}{T}\sum_t L_t \leq \alpha + \frac{\lambda_{T+1} - \lambda_1}{\eta T}$ [b] | realised only |
| `aci_window` | window CRC at level $\alpha_t$; $\alpha_{t+1} = \min(2\alpha, \alpha_t + \gamma(\alpha - L_t))$, $\gamma = 0.005$ | $\frac{1}{T}\sum_t L_t \leq \alpha + \frac{\alpha_1 - \alpha_{T+1}}{\gamma T}$ [c] | full + realised |

[a] The weighted-CRC bound $\alpha + \sum_i \tilde{w}_i \, d_{\text{TV}}(Z, Z^{(i)})$ [63, 64] cannot be computed. Items whose truth is unknown are left out of the window, which can bias it.
[b] The right-hand side is a certificate computed after the fact. It is bounded in advance only if $\lambda_t$ stays bounded, which holds for scores $\leq 1$ and $f \equiv 0$; then $\frac{1}{T}\sum_t L_t \leq \alpha + \frac{1 - \lambda_1 + \eta(1+d)}{\eta T}$.
[c] The upper clip at $2\alpha$ only lowers $\alpha_t$, so the bound stays valid. $\alpha_t$ stays bounded below because below the CRC feasibility level ($\approx 1/(N+1)$) nothing is committed and $\alpha_t$ rises.
All bounds count only the losses that the controller observes. Items whose feedback has not arrived are skipped; in our runs all feedback arrived after the delay, except in the "noticed errors only" runs, where a silent error is fed back as a correct commit.

Table 15 gives the online results on the fixed deployment splits (warm start on the clean calibration data, then the test data in file order, delay 0). Shuffling the test order changed the online rates by at most 0.007, and a feedback delay of 10 items by at most 0.009. The long-run bounds say nothing about short windows: the largest rate in a rolling window of 50 items was 0.06–0.08 (B2T '25 ours, split A), 0.10–0.12 (split B) and 0.12–0.18 (published) for the online methods, and 18% of the windows of `aci` on split A were above $\alpha$. When only noticed errors were fed back (delay 10), `aci` reached 0.0270 and 0.0527 (ours, splits A and B), 0.0502 (published, both splits), 0.0581 (sEMG cross-session) and 0.0563 (sEMG session-calibrated), while its certificate on the observed losses was at most 0.0509: silent errors look correct to the controller. The sEMG online order is participant by participant, not a time sequence.

**Table 15: Online re-calibration on the fixed deployment splits** (long-run test wrong-commit rate). Cert. / a priori: `aci` bound after the fact and in advance. sEMG cross-subject: no method committed.

| Data | `window` | `decay` | `aci` | `aci_window` | Cert. / a priori |
|---|---|---|---|---|---|
| B2T '25 ours, split A | 0.0245 | 0.0257 | 0.0270 | 0.0245 | 0.043 / 0.068 |
| B2T '25 ours, split B | 0.0478 | 0.0453 | 0.0527 | 0.0490 | 0.053 / 0.066 |
| B2T '25 published, split A | 0.0429 | 0.0392 | 0.0478 | 0.0453 | 0.048 / 0.059 |
| B2T '25 published, split B | 0.0441 | 0.0392 | 0.0478 | 0.0453 | 0.048 / 0.059 |
| sEMG cross-session | 0.0472 | 0.0381 | 0.0526 | 0.0490 | 0.053 / 0.081 |
| sEMG session-cal. | 0.0490 | 0.0454 | 0.0490 | 0.0490 | 0.049 / 0.072 |

**Table 16: Fixed deployment splits: clean vs reused calibration** (pure CRC; test sets). Reused: the full-fit pipeline calibrated on its own fit data. Clean CRC: fit-A pipeline, calibration data disjoint from all fitted parts (95% CI over days or participants, conditional on the calibration set). Bound: CRC calibration bound $(n\hat{R}_n(\hat{\lambda})+1)/(n+1)$; a value well below $\alpha$ means CRC was not binding. Exch. ref.: mean (5th–95th percentile) over 1,000 random splits of the pooled calibration and test items; $p = (k+1)/(B+1)$, one-sided; Holm: corrected over the six tests. sEMG cross-subject: no method committed. Online methods: Table 15.

| Data ($n_{\text{test}}$) | Reused | Clean CRC [CI] | Bound | Exch. ref. | $p$ | Holm | LTT, pure |
|---|---|---|---|---|---|---|---|
| B2T '25 ours, split A (816) | 0.0625 | 0.0245 [0.015, 0.035] | 0.0075 | 0.018 (0.014–0.022) | 0.012 | 0.072 | 0.0270 |
| B2T '25 ours, split B (816) | 0.0625 | 0.0600 [0.049, 0.071] | 0.0498 | 0.049 (0.029–0.071) | 0.21 | 1.0 | 0.0515 |
| B2T '25 published, split A (816) | 0.0490 | 0.0478 [0.029, 0.071] | 0.0498 | 0.049 (0.029–0.074) | 0.54 | 1.0 | 0.0294 |
| B2T '25 published, split B (816) | 0.0490 | 0.0490 [0.030, 0.073] | 0.0498 | 0.050 (0.031–0.074) | 0.50 | 1.0 | 0.0306 |
| sEMG cross-session (551) | 0.0635 | 0.0563 [0.039, 0.074] | 0.0486 | 0.045 (0.018–0.080) | 0.30 | 1.0 | 0.0309 |
| sEMG session-cal. (551) | 0.0417 | 0.0399 [0.025, 0.056] | 0.0486 | 0.047 (0.018–0.082) | 0.60 | 1.0 | 0.0218 |

# N What each calibration set was and was not used for

Both guarantees need the parts that define $s_i$ (temperatures, conformal sets, guard and policy thresholds) to be fitted on data disjoint from the calibration items. In the deployment protocol (Section 6) a fit part fits these and a disjoint calibration part only calibrates. Three exceptions remain. (1) Brain-to-Text '25: the word-decoder grid (LM weight, edit penalty, word bonus) was tuned on a random 400 of the 812 fit trials, so 187 and 189 of the 401 calibration trials of splits A and B were in that tuning set; re-tuning on fit-A only selected the same setting in both splits, so the reported rows are unchanged. (2) MEG: the decoder checkpoint was selected by validation CER, and the validation sentences include the 5 calibration sentences; this was not re-done (it needs GPU retraining). In addition, MEG instances are clustered by sentence, so instance-level CRC carries no guarantee for new sentences (Section 7.2). (3) sEMG session-calibrated: calibration and test logits come from two cross-fitted decoder runs, so their score functions differ by design; the calibration set had more eligible wrong items, which made CRC conservative there. LTT also needs i.i.d. sampling; our random splits sample without replacement from a finite pool, which makes its binomial test conservative.

# O Controls

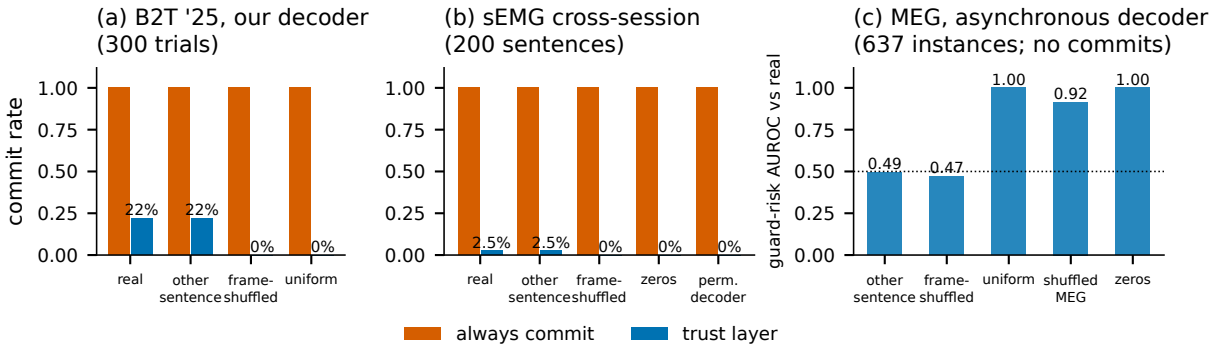



**Figure 7: Controls.** (a, b) Commit rate of always commit (orange) and the trust layer (blue) on real input, the intent-mismatch control (other sentence) and signal-destroying controls: (a) B2T '25, our decoder, 300 test trials; (b) sEMG cross-session, 200 test sentences. On controls other than "other sentence", any commit is wrong. (c) MEG, 637 instances: AUROC of the guard risk for separating each control from real input (dotted line: chance, 0.5); the MEG layer committed nothing.

# P How to state the guarantees

With LTT and the max rule (our primary rule): "if the calibration sentences and later sentences are i.i.d. from the same distribution, and the calibration sentences fitted no other part of the system, then, with probability at least 0.9 over the choice of the calibration sentences, the long-run probability that a sentence is committed wrongly is at most 0.05". With CRC: "under exchangeability and the same separation, averaged over the choice of the $n$ calibration sentences and over a new sentence, the probability that the new sentence is committed wrongly is at most 0.05"; a single deployment can exceed 0.05, and in our repeated splits a third or more did. With SGR: "with probability at least 0.9, at most a fraction $\beta$ of committed sentences are wrong". With adaptive online re-calibration: "for any sequence of sentences, the average observed wrong-commit rate over the first $T$ sentences is at most $0.05+(1-\lambda_1+\eta)/(\eta T)$" with no feedback delay; with delay $d$, replace $\eta$ in the numerator by $\eta(1+d)$. The unit of exchangeability matters: on MEG it is the sentence, and none of these statements applies to new sentences. All statements count only the errors that the system observes, so each should be reported with the measured silent-error rate.

# Q CRC and LTT on random splits

Random calibration/test splits of the test pool are exchangeable by construction, so they check the implementation, not the assumption. Table 17 gives CRC and LTT at four calibration sizes over 500 random splits: the mean wrong-commit rate of CRC stayed at or below $\alpha$ at every size, while single splits with 408 calibration trials exceeded 0.05 in 43% (our decoder) and 46% (published) of cases, partly from test-sampling noise (standard error about 0.011). Figure 8 shows a second set of 200 random splits with 100 and 408 calibration items (here $P_{>\alpha}$ was 0.445 and 0.415 for CRC at 408). Because the pool is the population of this experiment, the risk that LTT bounds, the rate of the chosen threshold on the whole pool, can be computed: over these 200 splits it exceeded $\alpha$ in 0.5% and 1.5% of LTT splits (within $\delta = 0.1$) and in 33% and 29% of CRC splits, which CRC does not promise to avoid. The fixed gate is the plug-in policy fitted on the fit data; it lies above $\alpha$ at both sizes.

**Small calibration sets.** With 20 calibration trials, CRC kept the mean valid but had a 90th percentile of 0.089 and never committed in 17% of splits; LTT cannot commit below 45 items.

A session-start block of the first 20 trials of each day gave CRC rates of 0.009 (ours) and 0.025 (published) at commit rates of 0.04 and 0.32, and 6 of 14 days never committed. On sEMG session-calibrated, with the first 20 test sentences of each participant as calibration, 9 of 18 participants never committed (one participant has only 11 test sentences and was left out).

**Table 17: CRC and LTT on random calibration/test splits of the test pool** ($\alpha = 0.05$, $\delta = 0.1$; 500 splits; the separate 200-split set of Figure 8 gives similar values). Wrong-commit: mean test rate; p90: its 90th percentile; $P_{>\alpha}$: share of splits with a test rate above $\alpha$; Commit: mean commit rate; Err. | commit: pooled over splits; Never: share of splits without a commit. MEG: pool of 637 instances.

| Data | Method, $n_{\text{cal}}$ | Wrong-commit | p90 | $P_{>\alpha}$ | Commit | Err. \| commit | Never |
|---|---|---|---|---|---|---|---|
| B2T '25, our decoder | CRC, 20 | 0.0308 | 0.089 | 0.22 | 0.123 | 0.25 | 0.17 |
| | CRC, 50 | 0.0342 | 0.071 | 0.21 | 0.146 | 0.23 | 0 |
| | CRC, 100 | 0.0456 | 0.075 | 0.38 | 0.186 | 0.25 | 0 |
| | CRC, 408 | 0.0479 | 0.069 | 0.43 | 0.192 | 0.25 | 0 |
| | LTT, 20 | 0 | 0 | 0 | 0 | — | 1.00 |
| | LTT, 50 | 0.0201 | 0.046 | 0.09 | 0.097 | 0.21 | 0 |
| | LTT, 408 | 0.0356 | 0.052 | 0.15 | 0.156 | 0.23 | 0 |
| B2T '25, published gate | CRC, 20 | 0.0308 | 0.073 | 0.23 | 0.345 | 0.09 | 0.03 |
| | CRC, 408 | 0.0495 | 0.071 | 0.46 | 0.484 | 0.10 | 0 |
| | LTT, 408 | 0.0355 | 0.054 | 0.15 | 0.429 | 0.08 | 0 |
| MEG, asynchronous | CRC, 408 | 0.0197 | 0.031 | 0 | 0.021 | 0.93 | 0 |
| | LTT, 408 | 0.0220 | 0.031 | 0 | 0.024 | 0.93 | 0 |

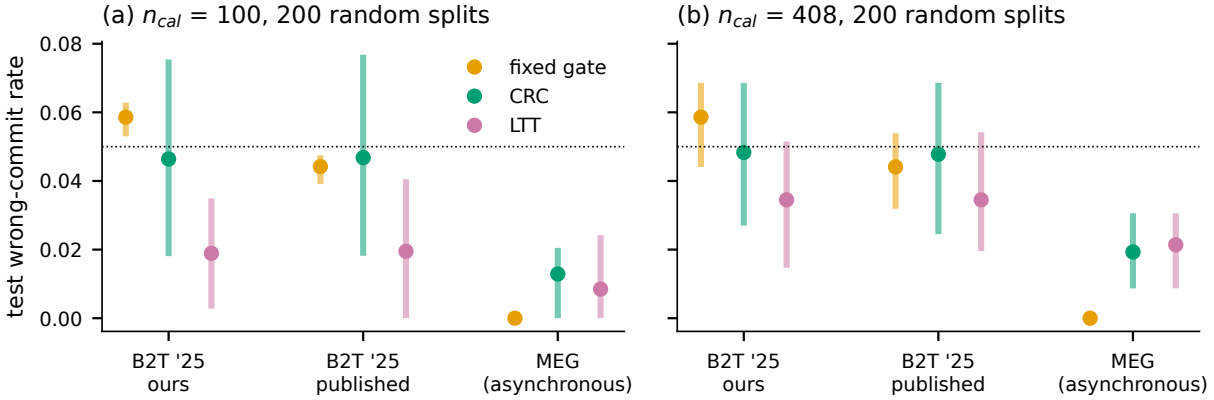



**Figure 8: CRC bounds the mean over splits; LTT bounds the calibration-conditional risk with probability $1 - \delta$.** Test wrong-commit rate over 200 random calibration/test splits with (a) 100 and (b) 408 calibration items. Dots: mean; bars: 5th to 90th percentile. Dotted line: $\alpha = 0.05$. LTT bounds the population risk, not the test rate of a finite split, which also contains test-sampling noise, so the 90th percentile of test rates need not be below $\alpha$. A test rate above $\alpha$ in a single split is not a violation of either guarantee.